Season 4 — The Repairs

Research document. Nineteen episodes. Season 3 ended with four failures and one diagnosis. This season is what happens when somebody believes the diagnosis and starts repairing, and it contains the first two things in the project that actually work.

> *Two anchors per concept, as in Seasons 2 and 3.* ***Character*** *for the audience,* ***clinical*** *for you. And from here the season carries a second spine underneath the first:* ***every repair in it is real, and most of them repair something that was not broken.***

The shape of the season

Five rounds, and for the first time some of them pass.

ROUND	THE REPAIR IT ATTEMPTED	RESULT
4	a fifth bucket, for the thing that actually separates Gospel from myth	**FAIL**, and its own controls destroy it
5	rebuild the null test whose script was lost	**UNSTABLE**, then identified another way
6	a null that matches the instrument for once	**PASS**, all four predictions
7	ask where in the text the words sit	**FAIL**, under-powered, and it says so
8	ask it again, better	**PASS**, and then takes most of it back
9	delete the word that was half a bucket	**inert**, which is its own kind of answer

The three things to hold onto before it starts

One. Season 3 diagnosed the problem as *whole-text rates cannot test claims about narrative structure*. Rounds 7 and 8 are the project taking its own diagnosis seriously, and they are the first rounds that ask **where** rather than **how much**.

Two. The season's successes are real and they are all in the same place. Watch for it, because nobody plans it and it is visible from episode 6 onwards.

Three. Season 3 asked you to write down whether the project was wrong or blind. **This season is where you find out**, and the answer is neither of the two things you were offered.

> **The refrain, and it changes register here.** A check that cannot fail prints reassurance. Seasons 2 and 3 watched that happen to *checks*. This season watches it happen to **repairs**, which is worse, because a repair carries the feeling of having done something.

Collecting Season 3

Nineteen episodes, four rounds, three failures and one pass that was thrown out for being vacuous. The closing diagnosis was exact and it was the season's whole payoff:

> **Whole-text rates cannot test claims about narrative structure.**

And underneath that sat a second thing, from Season 1 episode 11, which the project had not yet acted on. **The four stereotypes are a detector, never a discriminator.**

Those two words are about to carry the whole season, so spend a moment on them.

A **detector** answers "is the thing present here?" A **discriminator** answers "which of these two is which?" They sound like the same job and they are not, and an instrument can be excellent at the first while being structurally incapable of the second.

> A smoke detector is superb at finding smoke. **Point it at a kitchen and a house fire and it will report smoke in both**, correctly, at full volume, forever. No improvement to its sensitivity makes it better at telling you which room you are standing in.

Why the four buckets were never going to work

This is the sentence that should have been written in July and was written in August, and it is worth saying slowly because everything in this season follows from it.

Girard predicts that **the Passion scores high on all four stereotypes.** Not low. High. The Gospels describe a crisis, a crime, a marked victim and a collective killing, and they describe all of it deliberately.

So an instrument that measures the four stereotypes is measuring something the Gospels and the myths **share**. You can make it more sensitive, better tokenised, better de-noised, and every improvement makes it better at finding the thing both sides have.

	CRISIS	CRIMES	MARKS	VIOLENCE
a persecution myth	high	high	high	high
the Passion	**high**	**high**	**high**	**high**

Look at that table for longer than it deserves. There is no row here that separates anything. Four rounds were spent refining an instrument whose best possible output is a tie.

And notice this is not a complaint about the instrument being badly built. It was built well. Season 2 spent thirteen episodes making it honest, and every one of those decisions was correct. **A perfectly built detector pointed at a question about difference returns the same answer for both sides**, and it returns it with more and more confidence as you improve it.

> **This is the most expensive mistake in the project and it is not a mistake of execution.** Nobody ran the wrong test. Somebody ran an excellent test of the wrong thing, for three rounds, and the excellence is what made it hard to see.

The tell, in retrospect, is that the theory itself predicted the tie. It is in Girard, in print, and the project's own `FINDINGS.md` had written it down. **The information needed to stop rounds 2 and 3 was on disk before either of them ran.**

What Girard says the difference actually is

I See Satan Fall Like Lightning, page 3, which Season 1 episode 12 read twice and which is now doing a different job:

> "not merely the innocence of the victims versus their guilt, but the fact that, in mythology, **no one ever questions this guilt**. [...] In the Gospels, the revealing account of scapegoating emanates not from the unanimous crowd but from a dissenting few. [...] **In mythology no dissenting voice is ever heard.**"

Two things in that passage, and the project had been building an instrument for the wrong one.

The differentia is not innocence. Girard says so in the first clause. He is not claiming the mythical victim is guilty and the Gospel victim innocent. He is claiming that in myth **nobody argues.**

Work through why that distinction is the whole thing.

Take Oedipus. He really did kill his father and marry his mother, and the plague really does lift when he goes. **Girard is not saying Thebes got the facts wrong.** The city's account of itself is broadly accurate, and it is still a persecution text, because everyone in it agrees, and the agreement is what produces the expulsion.

Now take the Passion. The crowd's account is that a blasphemer was executed. **A Girardian reading does not primarily dispute that account. It notices who is allowed to speak.** Pilate says he finds no fault. The centurion says this man was righteous. One of the two crucified alongside says he has done nothing wrong.

	WHAT HAPPENED	WHO OBJECTS
Oedipus	a king is expelled and the plague lifts	**nobody**
the Passion	a man is executed and the city is quiet	**Pilate, a centurion, a thief**

> **The events are the same shape. The transcripts are not.** Girard's claim is about the second column, and every instrument in Seasons 2 and 3 measured the first.

The device: what a fifth bucket has to do that the first four never did

BUCKET	WHAT IT DETECTS	PRESENT IN MYTH	PRESENT IN GOSPEL
crisis	lost difference	yes	yes
crimes	the accusation	yes	yes
marks	the victim's selection	yes	yes
violence	the collective killing	yes	yes
dissent	**somebody breaking the unanimity**	**predicted absent**	**predicted present**

The fifth row is the first row in this project with two different words in it. That is the entire reason round 4 exists.

And notice what it costs. The first four buckets can be validated against texts everybody agrees are persecution myths. **The fifth cannot be validated against anything**, because if Girard is right there is no myth containing the thing it looks for.

That is a genuinely strange position to be in and it is worth stating carefully. Normally you check an instrument by pointing it at a known positive. A thermometer gets checked in boiling water. **There is no boiling water here.** The theory says the myth tier is empty of dissent, so any myth text you might use to calibrate the detector is, by hypothesis, a negative.

> So the fifth bucket goes into the field **uncalibrated**, and the only way to find out whether it can recognise dissent is to see what it does on the tier that is supposed to have some. **If the Gospels score high, that is either the instrument working or the instrument being wrong in a helpful direction, and nothing distinguishes those two from inside the result.**

Episode 8 is where that bill arrives.

The character anchor

Twelve Angry Men, and specifically the first vote.

Eleven guilty, one not guilty. Nothing about the evidence has changed. What has changed is that the room can no longer describe itself as unanimous, and the whole film is the consequence of that one broken count.

Now imagine the same film where juror eight votes guilty. **Same crime, same boy, same evidence, same verdict, and no story.** Every fact in the case is identical and the thing Girard is pointing at is simply absent.

> **That is the measurement.** Not whether the victim was innocent, which the film never settles. **Whether anybody in the room said so.**

The clinical anchor

A morbidity and mortality meeting where nobody speaks.

Every M&M has the same material: a bad outcome, a sequence of decisions, a room of people who were there. The facts of the case do not tell you which kind of meeting you are in.

What tells you is whether one person says the thing everybody is thinking. And a department where that never happens is not a department without errors. It is a department whose record of errors is written by the people who made them, which is Season 1 episode 12's point arriving in a corridor.

What would count against this

1. **Dissent being unmeasurable by vocabulary.** Juror eight's dissent is a *position in an argument*, not a word. A lexicon looks for words. **This is the objection that turns out to be right**, and it takes until episode 16 to see it properly.
2. **Girard saying the opposite elsewhere.** The claim rests on two sentences from one book. Season 1 episode 11 stated that ceiling and it has not moved.

3. **The Gospels being a special case for boring reasons.** They are four documents about one week, written by partisans of the victim. **Of course** they contain objection. A myth is not usually written by the victim's supporters, and no instrument in this season controls for that.

Handing off

Episode 2 builds the bucket, and the interesting part is not the words that go in. It is the five that were deliberately kept out, each one for a reason that made the test harder to pass.

Questions

1. If the Gospels score high on all four stereotypes, what exactly was round 2 proving when it passed?
2. **Before episode 2:** name three English words you would put in a dissent lexicon. Keep the list. You will want it in episode 9.
3. Is "nobody argues" a claim about the event, or about the document that survived it? Girard needs both. Which one can a text measure?

Collecting episode 1

The four buckets detect persecution and cannot discriminate, because Girard predicts the Passion scores high on all four. The differentia he actually names is **dissent**, and a fifth bucket is the first instrument in the project with a chance of telling the two apart.

The architecture, which is deliberately boring

`dissent.py`. Same shape as the four buckets in `lexicon.py`: explicit surface forms, no stemming, phrases matched against the joined token stream. Metric is **hits per 1000 words**, identical to everything before it.

Nothing new is invented here, and that is a decision. A new construct measured with a new method gives you two things to blame when it fails.

Spell that out, because it is a rule worth taking away from the whole series.

Suppose the round fails and you changed both the construct and the metric. Two explanations are now available, and **both of them are true statements about your own design**, so you can pick. Suppose instead you changed only the construct. A failure is now attributable, because the metric has a track record across three previous rounds on the same corpus.

> **Hold one thing constant so a failure has somewhere to land.** That is the entire reason a boring architecture is the right architecture here, and it is why round 4 does not also switch to a smarter statistic while it has the file open.

The split that turns out to matter more than the words

Two provenance tags, and this is the load-bearing choice in the whole round.

TAG	SOURCED TO	CONCEPTS
`core`	page 3, the questioning of guilt	innocent, guiltless, no fault, done nothing, unjust, false witness, protest, object, denounce, defend, plead
`unan`	page 121, the broken unanimity	division, dissent, some said, others said, withheld consent, spare him, minority

Every result is reported twice, with and without `unan`.

Read the two lists again and notice they are not the same kind of thing. `core` is the vocabulary of *saying the victim did not do it*. `unan` is the vocabulary of *the crowd not being one thing*. Girard's sentence contains both, which is why both are here.

> **Fix this in your mind now, because it decides the season.** `core` states Girard's actual claim. `unan` states a structural fact about crowd speech. **If they come apart, everything depends on which half is doing the work.**

Here is the difference in one pair of sentences, both of which could appear in a text.

	A SENTENCE THAT FIRES IT	WHAT IT COMMITS THE SPEAKER TO
`core`	"this man has done nothing wrong"	**a claim about the victim**
`unan`	"and there was a division among them"	**a fact about the crowd**

The first is somebody taking a side. The second is a narrator reporting that the room was not of one mind, **and it can be true of a crowd arguing about anything at all.**

Girard's page 3 sentence contains both because for him they are the same event seen from two angles: the dissenting few *are* the broken unanimity. **The question this season answers, without ever meaning to, is whether they stay together when you measure them.**

The device: five words that were kept out, and what each one cost

EXCLUDED	WHY	WHAT INCLUDING IT WOULD HAVE BOUGHT
`have mercy`	a healing petition in the Gospels, not dissent	a large Gospel-only boost
`rebuke`	used of demons and wind, not a collective verdict	frequent in all four Gospels
bare `witness`	Johannine testimony, not objection	John's single most repeated word
`condemn`	the crowd's act, not the objection to it	common across the tier
bare `refuse`	too generic to mean anything	scattered everywhere

Every one of those exclusions makes the hypothesis harder to confirm. Three of them are frequent Gospel words that would have inflated exactly the tier the prediction needs to win.

That is the shape you want and it is rare. A lexicon written to pass would keep `have mercy` and write a justification for it afterwards, and **the justification would not be false.** "Mercy is a plea on behalf of the afflicted" is a defensible sentence. It is also how you would build an instrument that cannot fail.

And there is no way to tell the two apart by reading the justification. That is the uncomfortable part. Both versions of this lexicon come with reasons, both sets of reasons are literate, and the only difference is the order in which the word and the reason were written.

> Which is why the exclusions are the evidence rather than the inclusions. **Anybody can justify what they kept. The question is what they threw away while it was still expensive to throw away**, and `have mercy` in the Gospels was expensive.

Season 2 episode 2 made the same argument from the other side, with the nine dead words that fire zero times. **Dead words prove the list was written before the texts. Costly exclusions prove it was written before the result.** Two different receipts for two different kinds of honesty.

The clinical anchor

Deciding your inclusion criteria before you see who walks through the door.

Every exclusion in a trial protocol narrows your population and costs you power. You write them anyway, in advance, because a criterion added after recruitment is not a criterion. It is a description of who you wanted to keep.

And the tell is always the same. Criteria written in advance tend to hurt you. Criteria written afterwards tend to help.

The character anchor

Ace Attorney, and the objection that gets overruled.

The game is built on one verb. You object, and most of the time the judge overrules you and the trial carries on, and the objection is still the thing that happened. It is in the record. Somebody said no out loud in a room organised around agreement.

Now picture a courtroom game with no objection button. Same evidence, same witnesses, same verdict, and nothing to press. That is the myth tier as Girard describes it, and the fifth bucket is a button-press counter.

And a button-press counter is a genuinely good instrument for this, because it does not require you to decide who was right. **It does not read the case. It counts interruptions**, which is exactly the level of ambition Girard's sentence licenses. He does not say the Gospel crowd was correct. He says somebody spoke.

> **And here is the flaw in the anchor, which is the flaw in the round.** Counting button presses tells you an objection was raised. **It does not tell you what was being objected to**, and episode 16 is that sentence collecting its debt.

What would count against this

1. **The tags being motivated rather than derived.** The pre-registration says so itself, in advance, and episode 4 reads the disclosure.
2. **Exclusions being easy to praise and impossible to audit.** You are shown five words that were kept out. **You are not shown the words that were never considered**, and there is no procedure anywhere in this project for finding those.
3. **"No stemming" cutting both ways.** It keeps the matching honest and it means `protested` and `protesting` are separate decisions, each one a chance to include the form that helps.

Handing off

Episode 3 picks the texts, and the problem is one this project has hit before in a smaller form: the Gospels are in a different English from everything they are being compared against.

Questions

1. Compare your three words from episode 1 against the two tag lists. Which tag did you write from, and did you notice you were choosing?
2. `have mercy` was excluded for a good reason. **Construct the good reason for including it**, then say how you would tell the two apart in advance.
3. Is there any way to audit words that were never considered? If there is not, what does that do to every lexicon in this project?

Collecting episode 2

The fifth bucket is built, tagged `core` and `unan`, with five frequent Gospel words deliberately excluded. The tags exist because the pre-registration disclosed that one half was at higher risk of being motivated than the other.

The problem, and it is the biggest threat to the round

The corpus is almost entirely **1900 to 1921 English**. Murray's Euripides, Butler's Homer, Miller's Loeb Seneca, Frazer's Loeb Apollodorus. Edwardian translators, working in a shared register, on Greek and Latin verse.

The Loeb Classical Library is worth naming, since it supplies a good part of the corpus.

It is a series begun in 1911. Each volume prints the Greek or Latin on the left page and an English translation facing it on the right, so the translation is meant to be read *against* the original rather than instead of it. **Loeb English is therefore literal and a little formal**, closer to a gloss than to literature.

That house style is a property of the corpus, and one of the reasons its texts sound alike.

The Gospels are not.

So any difference you measure between the tiers might be a difference between two kinds of English, and Season 3 episode 5 already put a number on how large that effect can be. Two translations of the *same play* differed by **17.2%**.

Understand what that 17.2% is, because the rest of the season leans on it. It is **the same text, the same events, the same characters**, measured twice through two translators, and the instrument disagreed with itself by a sixth. Nothing about Greek tragedy changed between those two runs. Only the English did.

> So 17.2% is the size of the wobble in the ruler. **Any difference smaller than that is not a fact about the texts**, and any comparison across two different Englishes inherits it before it starts.

And the Gospel comparison is worse than the *Agamemnon* one, because those two translators were contemporaries. Here one side of the comparison is Edwardian Loeb prose and the other is a Bible.

The device: the choice, and the alternative that was rejected

CANDIDATE	DATE	WHY
ASV	**1901**	**chosen.** Literal, public domain, inside the corpus's own window
KJV	1611	rejected: three centuries off register

Read that table as a **control**, not a preference. The ASV is not chosen for being better. It is chosen for being **the same age as the rest of the corpus**, so that if a difference survives, the difference is not a date.

This is worth pausing on because it inverts the usual instinct. If you were choosing a Bible to read, the KJV is the obvious answer on almost every literary ground, and most people would reach for it without thinking. **Here it is the wrong choice precisely because it is the famous one.**

The question is not "which English is best" but "which English removes a variable I cannot otherwise control". 1901 against 1900-to-1921 removes the date. 1611 against 1900-to-1921 adds three centuries of drift on top of everything else you are trying to measure, **and any difference you found would have two explanations with no way to separate them.**

> **And the choice is made before scoring and not revisited.** That sentence is doing more work than it looks. The KJV is also on disk. Round 6 runs the entire test again on it, and the only reason that is a control rather than a rescue is that the ASV was named first.

What a checksum is for

The pre-registration records the source and a `sha256` hash of the text file, and states: **if a future download has a different checksum, that is a changed sample and needs saying out loud.**

This is worth explaining properly because it is not obvious.

A hash is a short fingerprint of a file. Change one character anywhere in the four Gospels and the fingerprint changes completely. It is not encryption and it hides nothing. **It exists so that "I used the ASV" becomes a checkable claim rather than a memory.**

WHAT YOU WRITE DOWN	WHAT IT PROVES
"ASV, from openbible.com"	you intended a particular text
"Matthew 23,427 words"	roughly the right book arrived
`sha256 aecff150...`	**the exact bytes, forever, by anyone**

Read the middle row again, because it is the one people think is enough. A word count catches a truncated download, which is a real failure and one this project has already had: Season 2 found a text cut off at 46% by a fetcher bug. It does not catch a different edition, a different versification, or a file where somebody's script stripped the verse numbers.

The hash catches all of those, and it catches them **without anybody having to anticipate them**, which is the property that makes it worth the trouble. You do not have to guess how the sample might change. Any change at all shows up.

The character anchor

Dubs against subs, for the same episode.

Two people watch the same anime and argue about whether a character is rude. One watched the dub, one read the subs. Neither is lying and neither watched the same script, because a dub is written to fit mouth flaps and a sub is written to fit a line of text.

Now count rude words in each and publish the difference as a fact about the character. That is the register confound, and this is a genuinely different instance from

Season 3 episode 5's two subtitle tracks, because here the two versions are separated by **290 years** rather than by two translators in the same decade.

The clinical anchor

The same hormone measured on two different analysers.

One sample, split in half, run on two manufacturers' platforms. The numbers come back different, sometimes by a fifth, and **neither machine is faulty.** They use different antibodies, which bind slightly different parts of the molecule, so they are answering slightly different questions about the same blood.

Which is why a result from one platform cannot be read against a cut-off derived on the other, and why a patient followed across a lab change gets a trend line with a step in it that is not in the patient.

> **That is the translation problem exactly.** Murray and the ASV are two instruments measuring the same construct, and the difference between them is real, systematic, and nobody's error. Season 3 episode 5 measured the size of that step at **17.2%**, and this episode is about choosing the second instrument so the step does not land inside the comparison.

What would count against this

1. **The ASV being closer to the Greek than any myth translation is to its original.** A literal translation preserves source idiom. If the Greek New Testament repeats a division formula, a literal English version repeats it, and **that is a fact about Greek, not about dissent.** Nothing in this project tests it.
2. **One translation not being a control at all.** A control needs two. Round 6 supplies the second and the answer is not clean.
3. **The 800-word rule and the corpus count.** Season 2 episode 4 showed the corpus size has four different values in the record. That is unresolved here too, and episode 17 finally counts it.

Handing off

Episode 4 writes the predictions, six of them, and one is registered by somebody who says in advance that he expects it to fail.

Questions

1. The ASV was chosen for its **date** rather than its quality. **Name a case where choosing on date would be the wrong call**, and say what you would choose on instead.
2. A hash proves the bytes. **Name something about a corpus that a hash cannot prove**, and that this project would need.
3. If the ASV's phrasing comes from the Greek rather than from 1901 English, is the confound smaller or larger? Answer before episode 13.

Collecting episode 3

The texts are fixed, the ASV is chosen for its date rather than its quality, and the file is hashed so the sample is checkable by anyone. The register confound is named as the largest threat and it is not yet controlled.

Two pieces of vocabulary first, because the scorecard prints both

Mann-Whitney U, and it is much simpler than the name.

You have four Gospel scores and seventy-one myth scores. Line all seventy-five up together and ask one question: **how often does a Gospel text beat a myth text?** Compare every Gospel against every myth, one pair at a time, and count the wins. That count is **U**.

every possible pair	4 × 71 = **284**
Gospel wins all of them	U = 284
Gospel and myth evenly matched	U = **142**
observed	**U = 268**

> **So U = 268 means the Gospels won 268 of 284 head-to-head comparisons.** That is the whole test. No formula, no assumption about the shape of the data, just counting who beat whom.

And it is used here instead of comparing averages **because it does not care how big the wins are.** One text with a wild score cannot drag the result, which matters in a corpus where `orphic_hymns_2` exists.

One-sided, which Season 3 episode 9 defined and which now has to be cashed.

The prediction is directional, Gospel **above** myth, so the test only counts surprises going that way, and **the p-value is half what a two-sided test would report.** $p = 0.00034$ here is $p = 0.00068$ two-sided, and the result survives either reading.

> **Season 3 episode 9 also named the cost, and this is the round where the bill arrives.** A one-sided test returns nothing for a huge effect in the wrong direction. Episode 5 contains exactly such an effect, in the `core` half, where tragedy climbs above the Gospels, **and no test in this round is capable of registering it as a result.**

Six predictions

	PREDICTION	TEST
D1	median dissent rate: MYTH < TRAGEDY < GOSPEL	ordinal, strict
D2	GOSPEL above MYTH	Mann-Whitney, one-sided, $p < 0.05$
D3	TRAGEDY above MYTH	Mann-Whitney, one-sided, $p < 0.05$
D4	the separation beats a frequency-matched null	95% of 2000 draws
D5	*Prometheus Bound* in the top 3 of 23 tragedies	rank
D6	no myth text above the lowest Gospel	the universal negative, literal

D5, and why it is the bravest one in the project

`FINDINGS.md` recorded, in July, **before the dissent construct existed**, that *Prometheus Bound* is the one tragedy scoring 0.00 on the four-bucket MIN and is "the one tragedy that unambiguously sides with its victim". And it forbade the claim until tested.

So there is a note in the record saying *this play is different*, written for reasons that have nothing to do with the instrument now being built. **That is the closest thing this project ever has to an out-of-sample prediction**, and D5 cashes it.

Why that is stronger than it sounds: the July note was made by an instrument that **could not measure dissent at all.** It noticed *Prometheus Bound* because the four-bucket MIN came out at 0.00, which is a fact about persecution vocabulary, and then a human being read the play and wrote down that it sides with its victim.

> **Two independent routes are supposed to meet here.** A structural oddity in July, on a different instrument, for a different reason, pointing at the same play the dissent lexicon should rank at the top in August. **If they meet, that is real convergence. If they miss, one of the two readings was wrong** and the project finds out which.

D6, and the sentence that makes this round worth watching

The pre-registration says, in the document, before any scoring:

> **I expect D6 to fail**, and it is registered anyway. A lexical rate almost certainly cannot deliver a categorical separation across 71 texts. Stating the expectation in advance is the point: if D6 fails, the honest conclusion is that **the universal negative is not testable with this instrument**.

Sit with how strange that is. A prediction is registered by somebody who has written down that he thinks it will fail, and who has said in advance what its failure will and will not mean.

Why bother? Because of the alternative. If D6 is not registered and a myth turns up scoring above a Gospel, there are two available sentences afterwards and both are true: *that was never the claim*, or *Girard is refuted*. **Registering the expectation removes the choice.**

And it removes a second one, going the other way, which is the half people miss. Suppose no myth beats the Gospels. Without D6 registered, that is a spectacular headline: **the universal negative holds across seventy-one texts.** With the expectation on record, the same outcome reads as "a blunt instrument produced the blunt result it was expected to produce", which is worth much less and is correct.

> **A prediction registered with its expected outcome cannot be spent twice.** It has one meaning if it passes and one if it fails, and both were written before anyone knew which. That is the only condition under which a result and its interpretation are the same object.

The device: what a registered expectation buys you

	D6 NOT REGISTERED	D6 REGISTERED WITH "I EXPECT THIS TO FAIL"
myth beats Gospel	pick the reading afterwards	**the instrument is too blunt**, fixed in advance
no myth beats Gospel	claim the universal negative	**a lexical rate cannot show that**, fixed in advance

Both columns look identical from outside. The difference is entirely in when the sentence was written, which is the series' whole subject arriving in a single table.

Look at the top-left cell for a moment, because it is the one that describes most published work. *Pick the reading afterwards.* Nobody in that cell is lying. There genuinely are two honest readings of a myth outscoring a Gospel, and a competent author will choose the one that fits the rest of the paper, and a referee cannot see the choice being made.

The right-hand column costs something real, which is why it is rare. It throws away the author's judgement at exactly the moment the author knows most, and it does so in exchange for a property that cannot be demonstrated any other way.

The character anchor

Calling your shot in pool.

Nobody makes you say which pocket. The ball goes in and it counts, and the room saw it go in. Calling it converts a good outcome into a demonstrated intention, and **the cost is that you also convert a miss into a called miss.**

Somebody who never calls a pocket has a strictly better-looking record and has never once shown you they knew what they were doing.

The clinical anchor

A primary endpoint you suspect is under-powered, declared as primary anyway.

Everyone in the room knows the trial is unlikely to hit it. You could demote it to secondary and give yourself room. The reason not to is that a trial with a generously chosen primary produces a paper that says nothing, forever, and nobody can tell from the outside.

What would count against this

1. **Registering an expected failure being cheap theatre.** It costs nothing to predict failure and then fail. **The check is whether the same document also registers things it expects to pass**, which D1 to D4 are.
2. **D5 being weaker than it looks.** The July note came from the same author. It is out-of-sample for the *instrument*, not for the *person*.
3. **Six predictions being enough rope.** With six, some will pass by chance. The pre-registration answers this by naming D1 to D4 as primary and D5, D6 as additional, **before the run**, which is the only time that split means anything.

Handing off

Episode 5 runs it. Three of the four primary predictions pass, and the number on the front of the results file says the round fails.

Questions

1. Write down now: will D6 fail? And **does its failure tell you anything about Girard?**
2. If a registered expectation of failure is free, design a check that catches somebody using it as cover.
3. D5 is a July note being cashed in August. **What would make it worthless?**

EPISODE 5 — THREE OF FOUR, AND THE ROUND STILL FAILS

Collecting episode 4

Six predictions, four primary, one registered as expected-to-fail, one cashing a July note about *Prometheus Bound*. All fixed before scoring.

The numbers

TIER	N	MEDIAN RATE /1000	MEAN	HITS	WORDS
GOSPEL	4	**1.387**	1.349	112	83,180
TRAGEDY	23	0.667	0.795	236	320,887
MYTH	71	0.547	0.594	365	612,835

	RESULT	
D1 ordinal medians	0.547 < 0.667 < 1.387	**PASS**
D2 GOSPEL > MYTH	U=268, **p = 0.00034**	**PASS**
D3 TRAGEDY > MYTH	U=1027, p = 0.032	**PASS**
D4 frequency-matched null	591/2000 beat it, p = 0.296	**FAIL**
D5 *Prometheus Bound* top 3	**rank 13** of 23	**FAIL**
D6 no myth above lowest Gospel	**7 of 71** exceed it	**FAIL**

Read the top row before anything else

The ordinal series came out exactly as Girard predicts. Myth below tragedy below Gospel, in that order, on a construct sourced to two sentences of his, with the words fixed in advance and five frequent Gospel words deliberately excluded.

And the Gospel-versus-myth separation is at **p = 0.00034**, which is not a marginal number.

This is the best-looking result in the project so far, and it arrives in a file whose title says the round fails.

Take a second on p = 0.00034 so the rest of the episode lands. It means: if the Gospels were drawn from the same population as the myths, a separation this large would turn up about **three times in ten thousand**. That is not a marginal finding, it is not a borderline call, and nothing about the arithmetic is in dispute.

> **Every number in that scorecard is correct.** The round does not fail because somebody made a mistake in it. It fails because of two paragraphs written weeks earlier, by the same person, in the document that set it up.

Why 3 of 4 is not 3 of 4

Because the same document that registered the predictions also registered two sensitivity analyses, in advance, and **both of them break a passing prediction.**

S1, core only. Drop the `unan` tag, the half disclosed as higher-risk:

```
core only:   GOSPEL 0.402   TRAGEDY 0.503   MYTH 0.325
             ordinal BREAKS, tragedy now above Gospel
             GOSPEL > MYTH  p = 0.293      <- D2 gone
             TRAGEDY > MYTH p = 0.005      <- survives
```

The concepts that state Girard's actual claim carry none of the effect. Innocent, guiltless, no fault, unjust, false witness, protest, denounce, defend, plead. All of it, nothing. The contaminated half carries the entire result.

And look at what the ordinal series does when `unan` is removed. It does not weaken. **It inverts.** Tragedy climbs above the Gospels, which is the opposite of the prediction, on the words that state the prediction most directly.

That deserves naming rather than passing over. p = 0.293 is not "a smaller effect". It is **no effect**, and the tier ordering has gone backwards, and this is the half of the instrument built from the sentence the whole round was sourced to.

S2, epic excluded. Restrict MYTH to mythography:

```
MYTHOGRAPHY 0.674    EPIC 0.382    TRAGEDY 0.667
TRAGEDY > MYTHOGRAPHY  p = 0.388   <- D3 gone
```

Mythography scores above tragedy. D3 passed only because 37 epic texts drag the myth tier down. There is no myth-below-tragedy series. There is **epic below everything**, which is a fact about Homer's register.

This is the same lesson as S1 in a different costume, so state it once in general terms. **A tier is not a natural object.** MYTH here is 34 mythography texts plus 37 epic texts, and those two groups behave completely differently. A median taken across them describes neither, and it moves depending on how many of each you happen to have collected.

> So D3 was never really testing tragedy against myth. **It was testing tragedy against a mixture whose composition nobody chose deliberately**, and the mixture was 52% epic by accident of what was on Perseus.

The device: the same round, scored twice

	HEADLINE READING	WITH ITS OWN REGISTERED CONTROLS
D1 ordinal	PASS	breaks under S1
D2 Gospel > myth	PASS, p = 0.00034	p = 0.293 under S1
D3 tragedy > myth	PASS, p = 0.032	p = 0.388 under S2
D4 null	FAIL	FAIL
count	**3 of 4**	**nothing survives**

Both columns are honest descriptions of the same run. The left one is what a paper would print.

And notice there is no dishonest step available to point at. Publishing the left column requires no false sentence. You would report D1, D2 and D3 accurately, mention the sensitivity analyses in a limitations paragraph, and every word would be true. **Season 3 episode 1 called this the temptation that is not a lie**, and here it is again with better numbers behind it.

The difference between the columns is not honesty. **It is whether the controls were binding**, and a control is only binding if it was written before you knew you would need to survive it.

The clinical anchor

A subgroup analysis that was registered rather than discovered.

The usual scandal is a subgroup found afterwards. This is the opposite and it is worth naming as a distinct thing: **a subgroup declared in advance, which then destroys your main result.** Nobody makes you run it. The protocol does, and the protocol was written by you.

The character anchor

A speedrun with the timer still running through the load screens.

The run is genuinely fast. The player is genuinely good. And the category rules, written by the community in advance, say load times count, so the record is not a record. **Everybody can see the run was excellent. It still does not go on the board**, and the reason it does not is a rule nobody applied specially to this person.

What would count against this

1. **The sensitivity analyses being too harsh.** Dropping `unan` removes half a construct sourced to Girard's own page 121. **Make that argument at full strength.** Then notice that the pre-registration wrote S1 *because* it suspected that half, and it was right.
2. **Epic not being myth.** If epic is excluded on principle, D3 was mis-specified rather than failed, and mythography above tragedy is the real finding.
3. **p = 0.00034 being real.** It is. **The question is what it is a p-value about**, and episodes 6 and 7 are the two different answers.

Handing off

Episode 6 asks why D4 failed, and the answer is not that the instrument is weak. It is that the null tested the half of the instrument with no signal in it.

Questions

1. Which column of the device table would you have published? Answer honestly before episode 6.
2. **The `core` half carries nothing.** Give the innocent explanation, then the damaging one, and say what would distinguish them.
3. If epic drags the myth tier down, what does that say about every myth-tier number in Seasons 2 and 3?

Collecting episode 5

Three primary predictions passed and none survived its own registered controls. The `core` half carries nothing, the `unan` half carries everything, and D4, the null test, failed outright at p = 0.296.

Splitting the instrument in half explains everything

The lexicon has two kinds of entry: **unigrams** (single words like `unjust`) and **phrases** (`others said`, `let him go`). That distinction has been sitting in the instrument since it was built and nobody had any reason to care about it, because a hit is a hit and the metric adds them all up.

It turns out to be the most important fact about the lexicon. Score the tiers on each half separately:

HALF	GOSPEL MEAN	TRAGEDY MEAN	MYTH MEAN
unigrams	0.367	**0.639**	0.486
phrases	**0.982**	0.156	0.108

The unigram half runs backwards. On single words the Gospels score *lowest* of the three tiers. The observed Gospel-minus-myth separation on unigrams is **−0.075 per 1000**, a negative number.

The phrase half separates the tiers by roughly **nine times**.

Sit with how strange that is before moving on. The same lexicon, built in one sitting from one pair of sentences, contains two halves that **point in opposite directions**. One says myths contain more dissent language than the Gospels. The other says the Gospels contain nine times more.

> Added together they gave the ordinal series that passed D1, D2 and D3. **The headline result of round 4 is a sum of two numbers that disagree**, and the sum came out positive because one of them is much larger.

And now look at what the null actually did

The null matched **unigrams only**. 45 of the 64 unigrams present in the corpus. **All 53 phrases were ignored.**

> So D4 built 2000 random word-sets, compared them against **the half of the instrument whose separation is negative**, and found that 591 of them did better. **Of course they did.** They were beating a number below zero.

D4 did not fail because the instrument is weak. It failed because the null tested the half that carries no signal, and the entire effect lives in the phrase half, which has never been tested against a null at all.

The device: what each half was asked and answered

	TESTED BY D4	CARRIES THE EFFECT	VERDICT AVAILABLE
45 unigrams	**yes**	no, runs backwards	tested, and it fails
53 phrases	**no**	**yes, all of it**	**untested**

The two ticks are in different rows. That is the whole episode.

Why this is a defect and not a disaster

The results file says it plainly: *that is a defect in my test design and it is recorded as one.* And then it does the thing that matters.

> The corrected version, a phrase-aware null, is **round 6 with its own commit**, not a rerun folded into this one. **Re-running until something passes is the move the project exists to refuse.**

Notice the specific discipline. The fix is known, it is cheap, and it would very likely turn a failed round into a passing one. **And it goes in a different file with a different date**, because a round that gets fixed until it passes is not a round.

Why the date matters, stated plainly: if the corrected null were folded back into round 4, the published record would show **a round that passed**, and the fact that it took two attempts would exist nowhere. Anybody reading it later would see one pre-registration, one result, one pass.

> **The number of attempts is part of the result.** A prediction that passes on the second null is a different object from one that passes on the first, and the only thing preserving that difference is somebody's willingness to open a new file instead of editing an old one.

The character anchor

A metal detector swept over a beach, with the sensitivity set for coins.

It finds coins. It reports the beach as coin-rich. Somebody asks whether the readings could be random noise, so you run the obvious control: sweep a hundred random patches of sand and compare.

And the whole time, the thing actually buried there is a car door. The detector found it, the reading is in the log, and the control you designed never looked at that channel because you built the control around coins.

The clinical anchor

A troponin drawn once.

The single most informative thing about troponin is not the value. It is the **rise and fall** across serial draws. Draw it once and you have measured the half of the test that carries the least information, then treated a normal result as reassurance.

And the assay was fine. The blood was fine. The sampling design threw away the signal, and the printout looks exactly like a printout that means what you want it to mean.

What would count against this

1. **The phrase half being a translation idiom.** "Some said, others said" is how the ASV renders divided crowd speech. Whether that is Girard's broken unanimity or a reporting convention **is exactly what a whole-text rate cannot distinguish**, and episode 13 is the test.
2. **Unigrams running backwards being the real finding.** The Gospels score *lowest* on single dissent words. Nobody has explained that, and it is at least as interesting as the phrase result.
3. **Splitting after seeing the failure.** The unigram-phrase split was not pre-registered. It is a diagnosis of a failure, not a result, and it is reported as one.

Handing off

Episode 7 steps back, because this is the third time a null test has broken in this project, and the three breakages are the same mistake wearing three different costumes.

Questions

1. Season 3 episode 14 asked you to predict the other null failures. **Go and look at what you wrote.**
2. Is "the fix goes in the next round" a real discipline or a delay tactic? What would tell you which?
3. The unigram half runs backwards. Construct the explanation, then say what text you would go and read to check it.

Collecting episode 6

D4 failed because the null tested the 45 unigrams and ignored all 53 phrases, and the unigram half runs backwards. The effect lives entirely in the phrases, which have never been tested against anything.

That is the *proximate* cause. This episode is the other one, and it is the same sentence Season 3 ended on.

What Girard actually claims, stated as a location

Read the claim again with an eye on where things are, not how many there are.

> At the moment of collective violence, a voice separates from the crowd.

At the moment. Not somewhere in the document. The dissent has to be *near the killing*, because a dissent that arrives in a different chapter about a different subject is not breaking anybody's unanimity.

The device: John's three division hits, with their addresses

HIT	CHAPTER AND VERSE	WHAT IS BEING DISPUTED
`division`	7:43	who Jesus is
`division`	9:16	whether he keeps the sabbath
`division`	10:19	whether he is mad

Not one of them is near a killing. They are theological arguments in the middle of the book, and they are the highest-scoring hits in the highest-scoring text in the whole corpus.

Read what those three verses actually are. A crowd arguing about a man's identity. A crowd arguing about sabbath observance. A crowd arguing about whether he is out of his mind. **All three are disagreements, all three genuinely break a unanimity, and none of them is anybody objecting to a killing.**

So the instrument is not malfunctioning. It found division, and division is what it was told to find. **The claim it was built to test was about division at a particular moment**, and the word "moment" never made it into the code.

A whole-text rate counts three. It cannot count *where*, because **position is precisely the information a rate discards.** That is not a bug in this implementation. It is what a rate is.

Work the definition, because this is the transferable idea. A rate is hits divided by words. **Both of those numbers survive any rearrangement of the text.** Cut John into sentences, shuffle them into a random order, glue them back together, and the dissent rate is identical to four decimal places.

> **Any statistic that survives shuffling cannot be measuring structure**, and narrative structure is the only thing Girard's claim is about. That is a test you can run on a proposed metric in your head, before writing any code, and it takes about five seconds.

The same failure, one level up from round 3

ROUND	WHAT IT MEASURED	WHY IT COULD NOT TEST THE CLAIM
3	`peak_min`, invariant to word order	shuffle the text, same number
4	**dissent rate per 1000 words**	**shuffle the text, same number**

> **Two failed rounds, one cause.** Whole-text rates cannot test claims about narrative structure. Round 3 found it for the four buckets. Round 4 finds it for the fifth, having built an entirely new construct in between and not noticed that the metric was the thing that failed last time.

That is the finding of round 4 and it is worth more than a pass would have been, and it is also slightly embarrassing, which is why it is worth saying out loud rather than in a footnote.

The embarrassment is specific and worth naming exactly. Round 3 did not fail mysteriously. It failed for a documented reason, written up in `RESULTS-3.md`, in July, in a file the same person wrote. **The diagnosis was correct, recorded, and then not carried forward into the design of the next round.**

> That is a different failure from getting something wrong. **It is knowing something and not applying it**, which no amount of rigour inside a round can catch, because the round is internally impeccable. Rounds 7 and 8 are what it looks like when the lesson finally crosses the gap.

The character anchor

A security camera that logs how many times the door opened, and not when.

Forty-three openings on Tuesday. That is a real measurement, taken by a real device, and it is the same number whether the openings were spread across the working day or all forty-three happened in the ninety seconds after the safe was emptied.

And the only question anybody has is which of those two Tuesdays it was.

> Nobody would call that camera broken. It counts correctly. It was **installed to answer a question it structurally cannot answer**, and every improvement to its counting accuracy leaves the question exactly where it was.

The clinical anchor

An Apgar score without the timestamp.

Seven is reassuring at one minute and frightening at five. It is the same seven. The number carries almost no information until you know **when it was taken**, and a chart recording a bare seven has recorded something that looks like data and cannot be acted on.

What would count against this

1. **Position being measurable and this being fixable.** It is, and rounds 7 and 8 do it. **The objection is that they should have been round 4.** The information that a rate cannot locate an episode was already in `RESULTS-3.md`, in writing, in July.

2. **The three division hits being cherry-picked.** They are the three in the highest-scoring text. **Nobody has read the other 109 Gospel hits by location**, and the argument would be much stronger if somebody had.
3. **Girard not requiring co-location.** Season 1 episode 11 showed that his written rule is looser than the project's. **If dissent anywhere in the document counts, round 4 passed and this episode is wrong.** That reading is available and it is not obviously worse.

Handing off

Episode 8 reads the seven myth texts that beat the Gospels, by name, because the pre-registration committed to reading them rather than explaining them away.

Questions

1. Rewrite Girard's claim as a sentence about **distance**. How would you measure the thing you just wrote?
2. If position matters, what is the right unit? Words, sentences, scenes, or something else? Each gives a different answer.
3. **The objection in item 3 is serious.** Which reading of Girard makes the project's whole approach right, and which makes it an invention? Commit.

Collecting episode 7

Round 4 failed for the same reason round 3 did: a rate cannot locate anything, and Girard's claim is about location. Before leaving the round, there is a promise in the pre-registration to keep.

The commitment, written before the run

> **A myth text with a sustained dissenting voice falsifies Girard's universal negative outright.** If a myth ranks above the Gospels, that text gets read and reported by name. It is not explained away.

Seven do. D6 predicted none. Here are the top two, read.

`orphic_hymns_2`, the corpus maximum

5,812 words, rate **2.24**, the highest dissent score of any text in the corpus, Gospels included.

FORM	HITS	WHAT IT IS
`defend`	5	a petition addressed to a god
`blameless`	4	a standard Orphic epithet
`unjust`	3	hymnic vocabulary
`spare the`	1	a prayer

Every single one is a hymn addressed to a deity. "Blameless" is a fixed epithet, the way "swift-footed" is fixed to Achilles. "Defend" is a request, not an objection. There is no crowd, no victim and nobody disagreeing with anybody.

The Orphic hymns are liturgy. They are the ancient equivalent of a litany, a sequence of invocations addressed to gods one after another, and the vocabulary of innocence and protection saturates them **because that is what you say to a god**, not because anybody is questioning a verdict.

> So the text with the highest dissent score in the entire corpus is the text with the least narrative in it. **No crowd, no killing, and nothing to dissent from.** It beat four Gospels on an instrument built to find dissent at a lynching.

`ovid_metamorphoses_9`

9,633 words, rate 1.76. `plead` ×3, `object` ×3, `unjust` ×2, `divided` ×2. Ordinary rhetorical verbs inside speeches, and `divided` in its **physical** sense, which the lexicon cannot see.

And `hyginus_fabulae_2` is 613 words with **one hit**, which is a rate artifact on a short text and nothing else. The pre-registration's third sensitivity analysis exists to catch exactly that.

The device: the sentence that is available, and refused

> **The trap, named so it is not walked into.** The obvious next sentence is "so these are false positives, therefore Girard's universal negative survives." **That is the escape hatch and it is refused.**

Follow the logic, because it is the best reasoning in the round.

	WHAT THE INSPECTION SHOWS	WHAT IT LICENSES
the naive reading	the seven are not real dissent	Girard survives
the honest reading	**the instrument produces counterexamples that are not counterexamples**	**it cannot test a universal negative at all**

In either direction. An instrument that calls an Orphic epithet a dissenting voice could not have found a *real* dissenting voice and distinguished it from an epithet. So a myth containing genuine sustained dissent would have arrived looking exactly like `orphic_hymns_2`, and been dismissed exactly the same way.

Follow that through to the end, because it is the part that stings. Suppose there **is** a myth somewhere in the seventy-one with a genuine dissenting voice at the killing. It would appear in this results file as a text with a high dissent rate. Somebody would open it, read the hits, find that most were rhetorical or hymnic, and file it with the others.

> **The procedure that correctly dismissed six false positives would have dismissed the real one too**, using the same reasoning, in good faith. And nothing in the output distinguishes the run where that happened from the run where it did not.

> **Girard's claim is neither supported nor falsified here. It is untested.**

That sentence costs the project the most quotable thing it could have said, and it is the correct sentence.

Compare the two headlines that were available from identical data. *Seventy-one myths were tested against Girard's universal negative and none of them contains a genuine dissenting voice.* Against: *the instrument cannot tell an Orphic epithet from a dissenting voice, so nothing was tested.*

The first one is publishable and the second one is true, and the only thing standing between them is somebody deciding that inspecting his own false positives tells him about his instrument rather than about the world.

The character anchor

A plagiarism checker that flags the bibliography.

It returns 40% matched. You look, and it is the reference list, the standard methods sentence and three quotations that are correctly attributed. Nothing is plagiarised.

Now ask the question that matters: **would this tool have caught real plagiarism?** You cannot tell from this run. All you have learned is that it matches strings, and the essay you just cleared might contain a stolen paragraph that happens to be paraphrased.

Clearing the essay and trusting the tool are two different acts, and the inspection only supports one of them.

The clinical anchor

A blood culture growing coagulase-negative staph.

One bottle, skin flora, almost certainly contamination from the draw. The right call is usually to ignore it, and the right call is usually right.

And the reason it is only "usually" is the whole problem. That organism does cause real line infections. A lab that returns it constantly has taught you to discard exactly the result that would matter, and **you will discard the real one in the same voice you used for all the false ones.**

What would count against this

1. **Reading only the top two.** Seven exceeded the threshold. Two are inspected in detail. **Five are not**, and one of those five is where a real counterexample would be hiding.
2. **"Untested" being too generous.** It is also possible the instrument is simply bad and the honest word is "failed". The difference matters for what round 6 is allowed to claim.
3. **The epithet reading being an interpretation.** Somebody could argue that a hymn calling a god blameless *is* a culture rehearsing innocence-talk. That is not obviously wrong, and no procedure here settles it.

Handing off

Episode 9 leaves round 4 for something that has been outstanding since July: a number the project has been citing for three weeks, whose script does not exist.

Questions

1. Read the argument in the device table again. **Where exactly does "false positive" stop implying "the theory survives"?**
2. Five of seven were not inspected. Design the rule that says how many you must read, in advance.
3. Is `blameless` in a hymn evidence about a culture's relation to victims? Argue both, then say which way you would have to write it down first.

Collecting episode 8

The seven counterexamples were read and none of them is dissent, and the conclusion drawn was that the instrument cannot test a universal negative in either direction rather than that Girard survives.

The number this whole project has been standing on

From `RESULTS-2.md`, July, and it has appeared in Season 2 and Season 3 and in every summary of the project since:

```
Girard buckets:  +6.10 per 1000 words
random sets:     mean -0.16, sd 2.95
beat Girard:     8/2000     p = 0.0045
```

p = 0.0045. It is the strongest number the four-bucket track ever produced. It is the one result that survived Season 3 intact, and Season 3 episode 10 is built on it.

The script that produced it is not in the repository.

Not deleted in anger, not hidden. It was written, it was run, the numbers were copied into a results file, and the file containing the code was never committed. This is the single most common way a result becomes unreproducible and it almost never involves anybody doing anything wrong.

> **The result outlived its method**, and it kept being cited the whole time, because a number in a results file looks exactly the same whether or not the thing that produced it still exists.

What is actually missing, precisely

This matters because "not reproducible" is a phrase people use loosely.

PRESENT	ABSENT
the three numbers, committed in July	the script that computed them
the corpus they were computed on	the seed
the lexicon version	the matching procedure
a prose description of the method	how phrases were handled

Everything except the thing that turns the inputs into the output. The result is a claim with a method described in a sentence, which is the ordinary situation in most published science and is exactly what nobody notices until they try.

And the prose description is genuinely good. "2000 random word-sets drawn from the corpus and frequency-matched to the lexicon" is a real method, stated clearly, and somebody competent could implement it.

Two competent people implementing it will not get the same number. They will differ on whether the matching pool includes the target words, on whether frequency is matched exactly or within a band, on how phrases are handled, and on whether the corpus is the pre- or post-correction version.

> Four decisions, two ways each, is sixteen versions of "the method described in that sentence". **The sentence is not wrong. It is not a specification**, and nobody notices the difference until the code is gone.

The device: three ways to respond, and only one of them is available

RESPONSE	WHAT IT MEANS	WHY IT IS REFUSED
keep citing it	the number is true because it was written down	**this is what "trust me" looks like from inside**
withdraw it	no script, no claim	throws away a result that is probably right
rebuild it	**write a spec, then re-run**	**the only one that can come out either way**

> **And rebuilding has a trap in it that is worse than either alternative.** The reconstruction has free parameters the record does not fix, and **the target number is sitting there in view.** With four or five choices to make and a number to hit, a reconstruction can be tuned into agreement without anybody intending it.

Notice why the middle row is genuinely tempting rather than obviously wrong. Withdrawing $p = 0.0045$ costs the project its best number on the strength of a missing file, when the number is very probably correct. **Nobody wants to retract something for a bookkeeping reason.**

And notice why the top row is what almost everybody does. The number has been cited three times already. Continuing to cite it requires no decision at all, which is the property that makes it the default.

The clinical anchor

A result in the notes with no requesting clinician and no assay named.

"Ferritin 12" in the margin, undated, no lab, no reference range. It is probably right. Somebody probably wrote it down off a screen. And you cannot use it, because the number and the confidence in the number are two different objects, and the notes preserved only one of them.

The clinical instinct is to repeat the test. That instinct is the whole of round 5.

And notice the instinct is not scepticism about the number. Nobody thinks the ferritin was invented. **Repeating it is cheaper than establishing whether the first one can be trusted**, and the repeat produces something you can act on, which the original never will however true it happens to be.

The character anchor

A save file with no game disc.

The save is intact. It records a hundred hours, every item, the exact position on the map. It is genuinely your progress, and it is a record of a world you can no longer enter to check anything.

And you can rebuild the game from the same version and load the save, which is the honest move, and which will tell you whether the save was ever consistent with a real playthrough.

The failure mode is what makes the anchor exact. Reinstall a slightly different patch and the save might load into a world where your position is inside a wall, and you will not be able to tell whether the save was corrupt or the patch was wrong. **Two unknowns, one observation**, which is the position round 5 is about to be in with four free parameters and one target number.

What would count against this

1. **This being pedantry.** The number is almost certainly fine. The method is described in prose. Reconstructing it costs days. **Make that argument properly**, then notice it is available for every unreproducible result ever published, including the wrong ones.
2. **The reconstruction being worse than the original.** A rebuilt script written by somebody who knows the target is a different object from a script written blind. Episode 10 is entirely about containing that.
3. **Nothing depending on it.** If $p = 0.0045$ vanished, the four-bucket track was already abandoned. **Except that Season 2 and Season 3 both cite it**, and so does this series.

Handing off

Episode 10 writes the specification, and the interesting part is that it is written and committed *before* the reconstruction, with the verdict rule fixed in advance, precisely because the target number is known.

Questions

1. You have a number, a prose method and no code. **Name the smallest addition to the record that would have made this episode unnecessary.**
2. Is "probably fine" ever good enough? Say what it depends on.
3. **Before episode 10:** if you had to reconstruct it, how would you stop yourself tuning toward 0.0045? Write your procedure down.

Collecting episode 9

The strongest number in the four-bucket track has no script behind it. Rebuilding is the only response that can come out either way, and rebuilding toward a known target is how you get agreement without earning it.

The problem, stated exactly

You are going to write a script. You know it must produce **p = 0.0045**. You have four or five decisions to make that the record does not fix, and every one of them has a defensible answer in both directions.

Nobody has to cheat for this to go wrong. You make a choice, the number comes out at 0.31, you think "that cannot be right", you look again at your phrase handling, you find something arguable, you change it, and the number moves. Every step was a real thought about a real ambiguity.

And the asymmetry is the whole mechanism, so name it. **When the number comes out near 0.0045 you stop looking.** When it comes out at 0.31 you keep looking, and looking longer finds more, because there genuinely are more ambiguities to find.

> **The bias is not in any decision. It is in how long you search before accepting an answer**, and search effort is invisible in the final write-up. Nobody records how many times they re-read their own code, and a script that was checked once looks identical to one that was checked eleven times.

> **This is Season 2 episode 7 again in a new setting.** Five true statements, a defensible chain, and a result at the end of it that nobody would call cheating.

The device: the grid, and why it is the answer

Instead of choosing, **enumerate.** Every free parameter gets every plausible value, and the script runs all combinations.

FREE PARAMETER	VALUES
corpus state	2
matching procedure	2
frequency pool	2
phrase handling	**2**
seed	3
	48 configurations, 2000 draws each

The seed needs a sentence, because it is the row that looks like padding and is not. A computer's random numbers are not random. They come from a formula started off by one number, the **seed**, and the same seed always produces the same sequence. That is what makes a study with randomness in it reproducible at all: quote your seed and anybody re-running your script gets your exact draws back.

> So varying the seed three ways asks a specific question. **Would this result have looked different if the dice had fallen differently?** If p moves when only the seed changes, the finding was luck. Across all twenty-four unigrams-only configurations it does not move, and that is a real part of why the reconstruction is trusted.

You cannot tune a grid toward a target. There is no single answer to nudge. Either the configurations agree or they do not, and their disagreement is itself the finding.

That is the move worth stealing, and it generalises well beyond this project. **When you cannot justify a choice, do not make it.** Run every version and report the spread, and the ambiguity stops being a threat to the result and becomes a measured property of it.

It also changes what a failure means. A single reconstruction that disagrees with the target tells you almost nothing, because it might be your one bad choice. **Forty-eight reconstructions that disagree tell you the target is wrong**, and forty-eight that agree tell you the choices never mattered.

The verdict rule, fixed before the run

This is the part that makes it a test rather than an exercise.

OUTCOME	VERDICT
all 48 give $p < 0.05$ **and** the observed separation matches	**REPRODUCED**
the grid straddles 0.05	**UNSTABLE**, and **$p = 0.0045$ cannot be cited**

Plus one more sentence, which is the one doing the real work:

> **No configuration may be selected as the correct reconstruction after the fact.**

Without that, UNSTABLE is worthless. Twenty-four configurations agreeing with your target is an invitation, and the sentence closes it before anybody sees the grid.

Picture the version where it is missing. The grid comes back split. You look at the twenty-four that agree and notice they share a parameter setting. You then write a paragraph explaining why that setting is the principled one, **and the paragraph is good**, because there is always a principled-sounding case for a technical choice.

> **The rule does not stop you making that argument. It stops the argument counting as a reproduction**, which is a different and much narrower restriction, and it is the only one that can be enforced from outside.

The character anchor

Rebuilding a lost level from speedrun footage.

The video shows what the level did. It does not show the code. So you rebuild it, and you keep comparing your version against the tape, and every time it diverges you adjust.

Eventually your level matches the tape perfectly, and you have learned nothing, because you fitted it. The only version of this that teaches you anything is building fifty variants by rule and asking how many of them could have produced that tape.

And the fitted version feels like success the entire time. Every adjustment is a genuine improvement in the match, every session ends closer than it began, and the finished level is **indistinguishable from a correct reconstruction** by anybody who was not in the room. Including you, a month later.

The clinical anchor

Back-calculating a dose from a drug level.

The patient has a level. You know the drug, roughly the timing, not the exact dose or the exact time given. You can produce a dose that fits, and you can produce several, and the honest output is not a dose. **It is a range, plus a statement of which unknowns the range is most sensitive to.**

Reporting a single number here is not more precise. It is less honest.

What would count against this

1. **48 not being enough.** The grid covers the parameters somebody thought of. **A free parameter nobody listed is invisible to this method**, and there is no way to know how many there were.
2. **The plausible ranges being chosen by the same person.** Every value in that table is a judgement. A grid built from generous ranges is a grid that will straddle anything.
3. **UNSTABLE being unfalsifiable in practice.** If a straddle means "cannot be cited", and a straddle is likely for any complex reconstruction, **the rule may simply forbid all reconstruction.** Episode 11 is where that gets tested.

Handing off

Episode 11 runs the grid. The verdict is UNSTABLE, and then the grid does something nobody designed it to do.

Questions

1. Write the fourth row of the verdict table: what if 47 of 48 agree?
2. **The grid cannot see unlisted parameters.** Suggest a procedure that could detect one, even roughly.
3. Is a range with its sensitivities a weaker claim than a point estimate, or a stronger one? Answer before episode 11.

Collecting episode 10

A grid of 48 configurations, each of 2000 draws, with the verdict rule fixed in advance and a sentence forbidding after-the-fact selection of the right one.

The verdict, first

> **UNSTABLE.** 24 of 48 configurations give $p < 0.05$ and 24 give $p \geq 0.05$.

```
48 configurations.  p range 0.00050 to 0.54123   median 0.08296
observed separation 6.0518 to 6.1407      (committed +6.10)
null mean          -5.275 to +7.138       (committed -0.16)
null sd             2.461 to 13.089       (committed  2.95)
```

By the rule written in advance, **p = 0.0045 cannot be cited.** An exact split, twenty-four each way, is the worst outcome the grid could have produced.

Look at the range on that p-value before going further. **0.0005 to 0.541.** The same claim, the same corpus, the same lexicon, reconstructed forty-eight defensible ways, and the answer runs from "one in two thousand" to "a coin flip".

> That range is the honest measure of how much the missing script was carrying. **The prose description of the method left room for a thousand-fold difference in the result**, and nobody could have known that without building all forty-eight.

And then the grid does something it was not built to do

It does not scatter. It splits, cleanly, on one parameter.

ARM	N	P RANGE	NULL SD RANGE
`unigrams-only`	24	**0.00050 to 0.00700**	**2.46 to 3.58**
`phrase-aware`	24	**0.15892 to 0.54123**	10.48 to 13.09

Nothing depends on the corpus state, the matching procedure, the frequency pool or the seed. **Everything depends on phrase handling.**

Read the shape of that before the numbers. A grid that straddles a threshold could straddle it in two very different ways. It could **scatter**, with results smeared across the range and no pattern in which settings produced what, and that would mean the reconstruction is hopeless because the answer depends on everything at once.

Or it could **split**, cleanly, on one axis, with every configuration on one side of that axis agreeing with every other. That is a completely different situation. **A split says the record was ambiguous about exactly one thing**, and it names the thing.

Within the unigrams-only arm, twelve other combinations of corpus state, matching procedure, pool and seed move p between 0.0005 and 0.0070 and nothing else, which is remarkable stability across four free parameters.

The device: identifying the lost script without using its p-value

This is the best single move in the project and it deserves the full walk.

The committed record contains **three** numbers, not one. Two of them are independent of the p-value:

COMMITTED, JULY	UNIGRAMS-ONLY ARM	PHRASE-AWARE ARM	INSIDE?
null mean **−0.16**	−1.93 to +0.40	−5.28 to +7.14	**unigrams**
null sd **2.95**	2.46 to 3.58	10.48 to 13.09	**unigrams**
p **0.0045**	0.0005 to 0.0070	0.159 to 0.541	unigrams

> **The null mean and the null standard deviation were committed in July and could not have been fitted**, because nobody was reconstructing anything then. Both land inside the unigrams-only range and nowhere near the phrase-aware range, whose standard deviation is four times larger.

So the original matched unigrams only, and under that procedure the committed result reproduces. The identification does not use the number under dispute. It uses two bystander numbers that were never the point.

And that is why the after-the-fact-selection rule did not have to be broken. The arm was not chosen because it agrees on p. **It was chosen by two other quantities**, and its agreement on p is then a genuine reproduction.

The general principle is worth extracting, because it is one of the few genuinely reusable techniques in the whole series.

> **A record that commits several numbers can be identified by the ones nobody is arguing about.** The disputed number is the one under pressure, the one somebody might have massaged, the one selection acts on. The bystander numbers are safe precisely because they were never the point.

A null mean of −0.16 is a boring intermediate quantity that got printed because the script printed it. **Nobody in July had any reason to care what it was**, and that carelessness is exactly what makes it evidence in August.

The thing that makes it more than a rescue

The phrase-aware arm is not a valid null. It matches a phrase on its *first token's* frequency, so `with one voice`, which occurs 3 times, gets replaced by a word as common as `with`, which occurs 8,294 times. Median inflation **113×**. 206 phrase-occurrences replaced by 65,137.

So the arm that disagrees is broken, and it is broken in a way that makes it easier to beat, which is why it produces the high p-values. Replace a rare phrase with a common word and the random sets suddenly have far more material to work with, so more of them clear the bar.

And there is a second reason the unigrams-only arm is the right one for *this* particular reconstruction. **Round 2's lexicon is 99.1% unigram driven**: the phrases contribute 0.0526 of a 6.104 separation.

That figure is checkable rather than asserted, and the check is one subtraction. Score the corpus with the whole round-2 lexicon and note the separation. Score it again with the phrases switched off. **The difference is what the phrases were contributing**, and here it is 0.0526 out of 6.104, which is 0.9%.

> So the phrases were nearly irrelevant to round 2's result, and matching them badly could not have mattered much either way. **Unigram matching was the right procedure there, and it was catastrophically the wrong procedure in round 4**, where the phrases carry everything.

Same technique, correct in one round and disqualifying in the next, and **the thing that changed is not the technique but the instrument it was pointed at.**

The character anchor

Forty-eight people baking from the same half-written recipe.

The note says flour, butter, sugar, hot oven. It does not say how much butter. Twenty-four bakers read it as a shortcrust and twenty-four as a sponge, and the two groups produce two completely different things, cleanly split, with no in-between.

Now you find the original baker's note about the tin she used and how long it took to cool. Nothing about taste, nothing about the argument. And only one of the two groups' cakes could have needed that tin and that cooling time.

The clinical anchor

Identifying which analyser produced an old result, from its reference range.

The value is disputed. But the report also carries a range and a unit, and only one of the two machines in use that year used those. **The disputed number is not what identifies the source. The boring metadata around it is**, and metadata is harder to fake because nobody is arguing about it.

What would count against this

1. **This being a rescue with extra steps.** The verdict is UNSTABLE and the section after it argues for what survives. **The results file says so itself**, and states plainly that §4 is the exact move the rule was written to prevent. The defence is that the identification uses quantities the rule was not protecting.
2. **The phrase-aware arm being fixable rather than invalid.** If it is fixed properly, it becomes round 6's null, and **round 6 is where that argument goes from theory to result.**
3. **99.1% being convenient.** It is measured, not asserted, and it is checkable. But it is also exactly the number needed to make the reproduction admissible.

Handing off

Episode 12 collects three broken nulls, because the phrase-aware inflation is the third instance of the same design failure and the pattern is now impossible to miss.

Questions

1. Two bystander numbers identified the script. **What other bystander numbers should every result in this project be committing?**

2. If the phrase-aware arm had been the valid one, what would the verdict be?
3. UNSTABLE, then reproduced by another route. **What should a reader cite?** Write the sentence you would put in a paper.

Collecting episode 11

The grid split on phrase handling, the lost script was identified by its null mean and standard deviation rather than by its p-value, and the arm that disagreed turned out to be invalid because it inflated phrase frequencies 113×.

The pattern, laid out

Season 3 episode 14 asked you to predict what the other null failures would look like. Here they are, and they are the same mistake three times.

ROUND	THE NULL	WHAT IT DID WRONG	DIRECTION OF THE ERROR
3	Null B, word shuffle	favoured evenly spread sets over clumpy ones **by construction**	made passing easier for the wrong texts
4	D4, frequency-matched	matched **unigrams only**, ignored 53 phrases	tested the half with no signal
5	phrase-aware grid arm	matched a phrase on its **first token**	inflated rarity 113×, destroyed the comparison

> **Every one is a mismatch between what the statistic depends on and what the null holds constant.** That is the whole diagnosis, and it is one sentence.

Before working the definition, notice that all three were written by somebody who knew what a null test is for, and that none of the three is a coding bug. **Every one of them ran correctly and produced a number.** That is the property that makes this class of error dangerous: a broken null does not crash, it reports.

Working the definition, because this is the transferable part

A null answers one question: **what would this number look like if the effect were not there?** To build one you must destroy the effect and keep everything else.

Say that second sentence again slowly, because both halves are load-bearing and people only ever remember the first. **Destroy the effect** is the obvious part. **Keep everything else** is where all three failures live.

Everything else means: the length of the texts, the frequency profile of the vocabulary, the clumpiness of the words, the number of items being counted. If your null changes any of those while it is busy destroying the effect, you are no longer comparing your result against a world without the effect. **You are comparing it against a different world**, and the comparison means whatever that difference happens to mean.

So there are exactly two ways to fail.

FAILURE	WHAT HAPPENS	EXAMPLE
destroys too little	the effect survives into the null	round 3's Null B kept the clumping
destroys too much	the null is easier to beat than reality	round 5's arm made rare phrases common

Round 4's D4 is a third thing and it is the sneakiest. It did not destroy too much or too little. It operated on **the wrong object entirely**, leaving the phrases untouched and randomising only the unigrams.

Which means the effect was never in the null's way. The phrases sat there, identical, in every one of the 2000 draws, **contributing their full separation to both the observed value and the null**, so they cancelled out and the comparison was decided entirely by the unigrams that run backwards.

> **A null that leaves the effect in place is not a weak test. It is a test of something else**, and it will return a confident number about that something else, and the number goes in the results file under the heading you wrote before you knew.

The device: the question to ask before writing any null

ASK	ROUND 3	ROUND 4	ROUND 5
what does my statistic depend on?	clumping	phrases	phrase rarity
what does my null randomise?	order	unigrams	first tokens
do those match?	**no**	**no**	**no**

Three rounds, three no's, and the table takes ten seconds to fill in. Nobody filled it in, three times, and each time the failure was only visible afterwards.

Which raises the question the rest of this series keeps circling. If the check is that cheap and that effective, why did somebody who writes pre-registrations and declares falsifiers and discloses contamination not run it?

> **Because the null felt like the check rather than the thing being checked.** Every one of those rounds had a careful pre-registration protecting the *prediction*, and the null test sat outside that scrutiny, in the part of the work labelled "controls", which is the part everybody trusts.

That is worth carrying forward past this project entirely. **The apparatus you built to catch yourself is the part nobody audits**, and it is the part where an error is invisible, because its output is reassurance.

The character anchor

Testing a lock by pulling on the door.

The door does not open. The lock is reported as working. And the way that lock actually fails is a bump key, which never involves pulling the door at all.

The test was real. The door was genuinely pulled. **It exercised a property the lock does not fail on**, and the result was recorded as reassurance in a log that now reads as a history of a secure building.

> **A check that cannot fail prints reassurance**, and this is the refrain arriving in a place nobody looks for it: not in the measurement, in the control.

The clinical anchor

Three near-misses with one root cause.

Three separate incident forms, three different wards, three different drugs. Filed separately, each one gets a local fix and a tick. Read together, all three are the same pharmacy label that puts the dose where the frequency should be.

The system that files them separately cannot see it, and every individual investigation was competent.

What would count against this

1. **Hindsight.** Each null looked reasonable when written, and the pattern is only obvious with three instances. **Say that honestly**, then notice that the three-row table above could have been written before any of them.
2. **The pattern being a coincidence of one author.** Three nulls by the same person may share a blind spot rather than reveal a general law. That is a real limit on how far this generalises.
3. **Round 6 not being a fourth instance.** It is the first null built by asking the question in the device table first. **Whether that is a repair or luck is testable**, and episode 13 is the test.

Handing off

Episode 13 builds a null that matches the instrument, and for the first time in the project every prediction in a round passes.

Questions

1. Fill the device table in for a null you would design to test whether dissent sits **near** violence. What must it destroy, and what must it keep?
2. Is there a null design that is safe by construction, or is this permanently a matter of judgement?
3. **Three failures by one author.** How much should that reduce your confidence in the fourth attempt, honestly?

Collecting episode 12

Three broken nulls, one mistake, and a three-row table that would have caught all of them. Round 6 is the first null in the project built by filling that table in first.

What the statistic depends on

The effect lives in **53 phrases**. So the null must destroy the phrases and keep everything else, and "destroy" here means one thing precisely: **replace each phrase with a phrase of the same shape and the same rarity that has no dissent meaning.**

Not a word. Not a phrase matched on its first token. **A real n-gram, of the same word length, at the same corpus frequency.**

Work through why each of those three conditions is there, because each one blocks a specific way of cheating yourself.

Same word length, because a three-word phrase has fewer chances to match than a one-word token, purely as arithmetic. Swap `let him go` for a single word and the null gets easier to beat for reasons that have nothing to do with dissent.

Same frequency, because rare things cluster. A phrase occurring three times in a corpus will, by chance, sometimes land twice in the same book. Replace it with something common and that clustering disappears, which is round 5's 113× failure exactly.

A real n-gram, rather than a random string of words, because the replacement has to be the kind of thing that actually occurs in this English. **Otherwise you have destroyed the effect and the grammar at the same time.**

An n-gram is just a run of n words that actually sit next to each other in a text. Two words is a **bigram**, three is a **trigram**. In "he found no fault in him", `no fault` and `found no` are bigrams, and `found no fault` is a trigram. Nothing more technical than that.

So the inventory is built by sliding a two-word window and then a three-word window across the whole corpus and writing down what you see. **378,412 distinct bigrams and 808,967 distinct trigrams**, counted as *types* rather than occurrences, which is why the trigram number is larger: there are more ways to arrange three words than two.

> That inventory is the thing that makes the round-6 null possible at all. **You cannot swap `let him go` for a comparable phrase unless you have a list of every comparable phrase in the language you are working in**, and the corpus is the only place that list can honestly come from.

The device: how the null is actually built

	ROUND 4'S D4	ROUND 5'S BROKEN ARM	ROUND 6'S N1
what is replaced	45 unigrams	phrases, by first token	**all 53 phrases**
replaced with	random unigrams	a single common word	**a real n-gram of equal length**
matched on	frequency	first token's frequency	**the phrase's own frequency, ±1.5×**
drawn from	word list	word list	**378,412 bigrams, 808,967 trigrams**

That last row is the part that takes work. The corpus's own inventory of every two-word and three-word sequence, built so that `let him go` can be swapped for something that occurs about as often and means nothing.

The result

	PREDICTION	RESULT	
P1	full lexicon beats the phrase-aware null	14 of 2000, **p = 0.0075**	**PASS**
P2	**phrases alone** beat it	**0 of 2000**, p = 0.0005	**PASS, primary**
P3	KJV phrase rate within 25% of ASV	20.4%	**PASS**
P4	GOSPEL > MYTH survives the swap	p = 0.00064	**PASS**

Not one of 2000 draws beat the observed phrase separation.

And the falsifier that mattered most did not fire: **F3, more than 30% of phrases unmatchable, came out at 0 of 53.** Every phrase found a partner of the same length and rarity, so the null is not under-powered the way round 3's was.

F3 is worth a sentence on its own because it is the falsifier nobody would think to write. If a phrase is so rare that no comparable n-gram exists, the null quietly skips it, and **the skipped phrases are exactly the rare ones carrying the most signal.** A null that silently drops the hardest cases will pass everything.

> So F3 is a check on the check. It fired at 0.0%, which means the comparison was made across the whole instrument rather than the convenient part of it, and that is the difference between round 6 and round 3.

> **All four predictions passed. That had not happened since round 2.**

The number that says the unigrams really do run backwards

```
full lexicon   +0.8396
phrases only   +0.9184
```

Phrases alone separate the tiers better than the whole instrument does, because the unigram half is dragging it down. That is the round 4 diagnosis confirmed by an independent route.

And it puts the project in an awkward position it never quite names. **The correct move, on the evidence, is to delete half the lexicon.** The unigrams contribute nothing, they run backwards, and removing them improves every number.

Nobody does it, and the reason is good: a lexicon edited to improve its own result is not the lexicon that was registered, whatever the justification. **So the instrument keeps carrying a half that is known to be dead weight**, and every result in the season is reported with it in, which is the price of the rule.

The character anchor

A control group for a card trick.

You claim you can find someone's card. The obvious control is a second deck shuffled at random, which tests whether *any* deck would work. But if your trick depends on the card being a face card, a control that swaps in number cards is testing something else entirely.

The right control swaps in a different face card, of the same rank, from a deck of the same wear. **Same shape, same rarity, no meaning.** And then finding your card is a fact about you rather than about face cards.

The clinical anchor

Matching controls on the variable your outcome actually depends on.

A case-control study matched on age and sex, where the outcome depends on renal function, has produced beautifully matched groups and answered nothing. **The matching was real work**, and it was done on the variables that were easy to extract rather than the ones the disease runs on.

What would count against this

1. **The n-gram inventory containing dissent phrases by accident.** 378,412 bigrams drawn from a corpus that includes the Gospels. **Some replacements will mean something**, and nobody has checked how many.
2. **The 1.5× frequency band being a free parameter.** Widen it and the null gets easier; narrow it and matches fail. It was fixed in advance, which is what makes it a parameter rather than a dial.
3. **P2 passing at 0 of 2000 being suspiciously clean.** A p of 0.0005 is the floor of a 2000-draw test. It means "we ran out of resolution", not "the effect is enormous", and it should be quoted that way.

Handing off

Episode 14 runs the whole thing again on a different translation, which is the test round 4 could not run and the one the project's own notes called the largest threat to the entire construct.

Questions

1. Why is a phrase's own frequency the right matching variable, and its first token's the wrong one? Say it in one sentence.
2. **P2 hit the resolution floor.** How would you find out whether the effect is large or merely non-zero?
3. Some random n-grams will carry meaning. Does that make the test conservative or anti-conservative? Think before answering.

Collecting episode 13

A null matched to the actual instrument, phrase for phrase, at 0 of 2000. The phrase half is not an artifact of rare n-grams. What it might still be is an artifact of **one translator**.

The threat, stated properly

Round 4's whole result was carried by the "some said, others said" family. That is how the **ASV** renders divided crowd speech.

A single translator's habit would produce exactly this result. One man's preference for a reporting formula, applied across four books, would put a cluster of identical phrases in one tier and nowhere else, and every null in the world would confirm it, because the phrases really are there.

This is the objection that no amount of statistical care can answer, and it is worth being clear about why. Round 6's phrase-aware null asks *are these phrases unusually concentrated in the Gospel tier?* The answer is yes, overwhelmingly. **A translator's habit produces exactly that yes.**

> Every null in the world tests whether the pattern is real. **None of them can test where the pattern came from.** For that you need a second measurement of the same thing through a different instrument, which in this case means a different translator.

So you need a second translation, made by different people, in a different century, from the same Greek.

The device: the same four books, two Englishes

GOSPEL	ASV /1000	KJV /1000	DELTA
Matthew	0.724	0.674	−6.9%
Mark	1.141	0.856	**−25.0%**
Luke	0.854	0.731	−14.4%
John	1.207	1.255	+3.9%
median	**0.998**	**0.794**	**−20.4%**

Inside the 25% bound, well inside the 40% falsifier.

Those bounds need explaining or the pass means nothing. 25% was the prediction and 40% was the falsifier, both fixed before the KJV was scored. The gap between them is deliberate: **there is a band where the result is disappointing but survivable, and a band where the whole construct dies.**

And the comparison that gives the number meaning is Season 3 episode 5's. Two *Agamemnon* translations, measured on a completely different construct, differed by **17.2%**, and the project called that its most load-bearing control.

> Those two translators were contemporaries working in the same idiom. **These two are 290 years apart** and land at 20.4%, barely three points worse. That comparison is what turns a bare percentage into evidence.

> **So the formula is not the ASV's private habit.** The 1611 translators, working independently in a different English, put comparable amounts of it in the same four texts. The most serious available explanation of round 4's result is now substantially weakened.

And it is a pass, not a clean bill of health

Mark drops 25.0% on its own, exactly at the bound. The median moves 20.4% in a single direction, downward. Any future round adding a third translation should expect movement of this size, and **nobody should quote a Gospel dissent rate to three decimals again.**

The direction is the part worth noticing. If translation were pure noise, the four deltas would scatter around zero. **Three of the four go down and the median goes down by a fifth**, which is not noise, it is the KJV systematically using less of this vocabulary.

So the honest reading is that translation contributes a real, measurable, one-directional component to the score, roughly a fifth of it, **on top of whatever the texts are doing.** The effect survives that. It is not clean of it.

The section that matters more than either pass

Round 4's sensitivity analyses were rerun on the KJV.

```
S1  core only, KJV GOSPEL > MYTH ............... p = 0.254
S2  epic excluded, KJV GOSPEL > MYTHOGRAPHY .... p = 0.011
```

Round 4 found that dropping `unan` took the Gospel result from p = 0.0003 to p = 0.293. **On the KJV it goes to 0.254. The finding is unchanged.**

> ***Everything round 6 established is about the `unan` half**, the half disclosed in advance as at higher risk of being motivated, because the author already knew John repeats a division formula before the lexicon was written.*
>
> *The `core` concepts, the ones that state Girard's actual claim, carry **none of the effect, in either translation.** What survives two nulls and a translation swap is division, some-said, not-all, spare-him, withheld consent.*

So the honest statement is narrow. A cluster of crowd-speech phrases is robustly more frequent in the Gospels than in myth, and it is not a translation artifact. **Whether that cluster is Girard's broken unanimity is a separate question, and nothing here answers it.**

The character anchor

Two different subtitlers agreeing that a character says "obviously" a lot.

If only one had noticed it, it is a translator's tic. Both noticing it, working separately, means the tic is in the original.

And it still does not tell you the character is arrogant. It tells you the word is there. The inference from a verbal habit to a character trait is a completely separate step, and it is the step everybody skips.

The clinical anchor

A finding confirmed on a second machine.

The lesion is on the CT and on the MRI. It is real, it is not a reconstruction artifact, and two different physical principles agree.

And you still do not know what it is. Confirming existence and identifying nature are different investigations, and a department that treats the second scan as answering the second question will operate on the wrong thing with excellent imaging.

What would count against this

1. **Both translations descending from the same Greek idiom.** If the Greek repeats a division formula, both Englishes will, and **the finding is about Koine Greek rather than about dissent.** Nothing in this project tests it, and episode 3 flagged it.
2. **The KJV not being independent.** The ASV is a revision in a line that includes the KJV. They are not two unrelated attempts.
3. **20.4% being close to the bound.** A bound set at 20% instead of 25% would have failed this. The bound was fixed in advance, which is the only thing that makes the pass meaningful.

Handing off

Episode 15 asks the question the project has been avoiding since Season 3: not how much dissent vocabulary a text has, but **where it sits.**

Questions

1. Both translations agree. **List everything that still does not follow.**
2. Is the ASV-KJV pair a genuine independent check, given their shared ancestry? What pair would be better?
3. **The `core` half has now failed in two translations.** At what point does a half of an instrument get deleted, and what would you lose?

Collecting episode 14

Two nulls and a translation swap, all passed, and all of it belonging to the `unan` half. The construct is real. Nobody has yet shown it is Girardian.

Now the project does the thing Season 3 said it must: it stops asking **how much** and starts asking **where**.

The measurement

For every dissent hit, how far is it from the nearest violence hit? Compare against a null that shuffles positions. If Girard is right, dissent sits **closer to violence than chance**, because a voice breaking the unanimity does it at the killing.

That is the first metric in nine rounds that would change if you shuffled the text, and that alone makes it a different kind of instrument. Rounds 2, 3 and 4 all measured quantities that survive rearrangement. **This one is destroyed by it**, which is exactly the property Season 3 said was missing.

The violence bucket, which the four-bucket track built and which Season 4 episode 1 declared useless as a discriminator, turns out to be essential here. **It is not being used to classify anything.** It is being used as a landmark, to say where in the text the killing is, and that is a job it can actually do.

The falsifier that fires before any prediction is read

Season 3 episode 15 built something after round 3 failed for low power: a **pre-declared power check**, so a round can be reported as *"we could not have seen it"* rather than *"it is not there"*.

This is the first round where it fires, and it fires inside the script, before any prediction is evaluated.

> **M3 FIRED.** Median z across all 66 included texts is **−0.012**, inside the ±0.2 band fixed in advance. By the pre-declared rule the round is reported as **under-powered**, not as an absence of co-location.

Read that number. **−0.012 is nothing.** The statistic is sitting on top of its own null in every direction. That is not a negative result, it is a measurement with no resolution.

The distinction between those two is the single most useful thing in this episode, so put it in plain terms. **A negative result says the effect is not there. An under-powered result says this experiment could not have seen it if it were.** They look identical in the output. Both produce a p-value above 0.05 and a disappointed author.

> And they license completely different next steps. A negative result means stop or change the theory. **An under-powered one means build a better instrument and come back**, which is what round 8 does, and it works.

The device: three ways a round can end, and only one was available before

VERDICT	WHAT IT CLAIMS	ROUND 3 COULD SAY IT?
the effect is there	positive	yes
the effect is not there	**negative, and citable**	**yes, wrongly**
we could not have seen it	**no information**	**no**

Round 3 had no way to say the third thing, so it said the second, and `RESULTS-3.md` recorded that its power problem "was foreseeable before running it and was not foreseen." Round 7 foresaw it, gave it a number, and the number fired.

And a second falsifier fired, which kills the round independently

M1 FIRED. Pass rates by tier:

TIER	PASS RATE
GOSPEL	**0.0%**
TRAGEDY	4.5%
MYTH	**7.5%**

The myth tier passes at a higher rate than the Gospel tier. Whatever the statistic is detecting, it is not tier-specific, and the ordinal prediction is dead however the rest landed.

	PREDICTION	RESULT
L1	3 of 4 Gospels below their null	**FAIL, 0 of 4**
L2	z ordered GOSPEL < TRAGEDY < MYTH	**FAIL** (+0.276, +0.665, +0.326)
L3	the *Bacchae* below its null	**FAIL**, p = 0.888
L4	more than 5% of texts pass	pass, **vacuously**

Every mean z is positive, and positive means dissent sits *farther* from violence than chance places it. In all three tiers.

All three going the same wrong way is itself informative, and it points at the null rather than at the texts. If dissent genuinely sat farther from violence in every genre of Greek and Christian literature, that would be a remarkable universal fact about narrative. **It is much more likely that the null is misplacing the baseline**, and round 8 finds out that it was: it shuffled the violence as well as the dissent, so the comparison drifted.

That is the fourth null-design failure, arriving one episode after the episode that catalogued three.

The prediction that was written down first, and came true

`PREREGISTRATION-7.md` stated before scoring that L1 was expected to fail, because round 4 had already recorded that John's three `division` hits sit at 7:43, 9:16 and 10:19, chapters away from the passion.

John comes out at p = 0.876. That was written down first, and it is the second time in the season that a disclosed expectation has been paid.

The character anchor

A photo taken in a dark room with a phone.

You look at it and there is nothing there. Two conclusions are available: the room was empty, or the exposure was too short. **The photograph looks identical in both cases**, and the only way to tell is to check what the camera could have picked up, which is a question about the camera and not the room.

M3 is checking the camera. It is the only reason the answer here is "no photograph" rather than "empty room".

The clinical anchor

A test whose confidence interval crosses everything.

The result is negative and the interval runs from a 60% reduction to a 90% increase. Reporting that as "no effect" is a straightforward misreading, and it is the most common one in the literature.

The scan was done. The number came back. And the study never had the resolution to answer its own question, which is a fact about the design and was knowable before anybody was recruited.

What would count against this

1. **The power check being set generously.** A ±0.2 band chosen wide enough will fire on everything and excuse every failure. **It was fixed in advance**, and whether 0.2 was honest is checkable by asking what band would have let this round report a negative.
2. **Under-powered being a comfortable place to land.** It costs nothing and sounds rigorous. The test is whether the project ever reports a plain negative again.
3. **M1 being the real story.** Myth passing more than Gospel is not a power problem. **That is a result, and it points away from Girard**, and the under-powered headline slightly buries it.

Handing off

Episode 16 fixes the null and asks again, and the answer is the first positive localization result in eight rounds.

Questions

1. What band would you have set for M3? Justify it without looking at the result.
2. **M1 and M3 both fired.** Which one should lead the results file? They point different ways.
3. Season 3 episode 15 said the power falsifier was "gameable by being set generously". **Was it?** Say what evidence would settle it.

Collecting episode 15

Round 7 was under-powered by its own pre-declared check, and its statistic was not tier-specific. The measurement was right in spirit and wrong in construction, because the null moved the violence as well as the dissent.

The fix, and it is one sentence

Hold the violence fixed. Move only the dissent.

Round 8 asks, for every dissent hit, what fraction of the text's positions lie at least as close to a violence hit. Call it **U**. Under random placement U is **exactly 0.5**, whatever the text length and however clumped the violence is.

Below 0.5 is Girard's prediction.

That is a better statistic than round 7's because it cannot be fooled by texts where violence is everywhere. The baseline is fixed by arithmetic rather than by a simulation that has to be trusted.

Take that claim apart, because "0.5 by arithmetic" is doing a lot of work. Pick any position in a text at random. Ask what fraction of all positions are at least as close to a violence hit as that one is. **Averaged over random positions, that fraction is 0.5, always**, for the same reason a randomly chosen person is above the median height half the time.

> The clumping of the violence does not matter. The length does not matter. **A baseline that comes out of the definition cannot be got wrong by a simulation**, which is precisely how rounds 3, 4, 5 and 7 all got theirs wrong.

The result

TIER	TEXTS	HITS	U	NULL MEAN	P
GOSPEL	4	112	**0.4171**	0.5018	**0.0015**
TRAGEDY	23	235	0.4838	0.5025	0.161
MYTH	50	334	0.4832	0.5018	0.124

> **In the four Gospels, dissent vocabulary sits closer to violence vocabulary than chance places it, at p = 0.0015.** That is the first positive localization result in eight rounds, and it is the shape of the claim Girard actually makes rather than the frequency claim rounds 2 and 4 tested.

And then the round takes most of it back

	PREDICTION	RESULT
U1	GOSPEL U below 0.5	**PASS**, p = 0.0015
U2	U ordered GOSPEL < TRAGEDY < MYTH	**FAIL**, 0.4171 / 0.4838 / 0.4832
U3	the *Bacchae* below 0.5	**FAIL**, p = 0.414
U4	gap ≥ 0.05 **and** p < 0.05	**FAIL**, gap +0.0661 but p = 0.123

U2 fails because tragedy is not intermediate. Tragedy 0.4838 and myth 0.4832 are the same number to three decimals. There is no ordinal series. There is **a Gospel effect and a flat background.**

U4 is the instructive one. The gap between myth and Gospel is +0.0661, which clears the margin the pre-registration demanded. But permuting tier labels gives p = 0.123, so **the gap is real in size and not distinguishable from chance.** Four Gospel texts against fifty myth texts gives a between-text permutation very little to work with, and that was foreseeable.

The falsifier that fired, and it is the season's spine

N3 FIRED. Run the whole thing on the `core` half alone:

```
full lexicon:  GOSPEL 0.4171  <  MYTH 0.4832  <  TRAGEDY 0.4838
core lexicon:  TRAGEDY 0.4838 <  MYTH 0.5053  <  GOSPEL 0.5143
```

The Gospels go from lowest to highest. On the concepts that state Girard's actual claim, the Gospels show no co-location at all, and if anything sit slightly *farther* from violence than chance.

Read the two orderings again as a pair, because a reversal is a much stronger signal than a weakening. If `core` carried a smaller version of the same effect, the sensible conclusion would be that both halves measure dissent and one is noisier. **It does not carry a smaller version. It carries the opposite.**

So the two halves of a single lexicon, built in one sitting from one pair of sentences, are not measuring the same construct at all. **They were never one instrument**, and every result in this season that pooled them was averaging two different things.

> **In the headline, as the pre-registration demanded:** what is localized near violence in the Gospels is **crowd-speech vocabulary, not the vocabulary of questioning a victim's guilt.**

This is the fourth consecutive round to find the same split. Round 4's S1, round 6's S1 on both translations, round 7's core sensitivity, and now round 8. **The `core` half has never carried anything, on any measure, in any round.**

The device: the season's real pattern, finally visible

ROUND	WHAT PASSED	WHICH HALF
4	D1, D2, D3	`unan`
6	P1, P2, P3, P4	`unan`
8	U1	`unan`
4, 6, 7, 8	**nothing**	**`core`**

Everything this season succeeded at belongs to the contaminated half. That is the thing episode 1 said to watch for.

And the coding error, reported by the author

`score8.py` printed "core order matches full lexicon? yes" and **did not flag the falsifier.** The check as written did not implement the check as designed. It was caught by reading the numbers, not by the script.

A falsifier that cannot fire is a check that cannot fail, and it happened here, in the round that needed it most, to somebody who had spent seven rounds writing about exactly this.

The character anchor

A smoke detector wired to the wrong circuit.

It has a light, it beeps when tested, and it has been on the ceiling for a year. The test button works because the test button is wired to the speaker. **Nothing connects the sensor to anything**, and the annual test confirmed the part that was never in doubt.

The clinical anchor

An alarm limit set outside the physiological range.

The monitor will alarm at a systolic of 20. It is on, it is configured, it is documented as configured. **It will never fire on a live patient**, and the chart shows a night of unbroken normal observations.

What would count against this

1. **U1 being the only pass out of four.** One prediction in four, and the three that failed include the ordinal series the whole construct was built on.
2. **Four Gospel texts.** Every Gospel number in this season rests on n = 4, and U4 failed specifically because of it. **The effect may be real and the design cannot show it**, which is round 7's lesson recurring.
3. **`unan` being Girardian after all.** "Some said, others said" *is* a crowd failing to be one voice. **That is exactly what Girard says breaks.** The objection to calling it contaminated is stronger than this season admits, and Season 7 has to settle it.

Handing off

Episode 17 goes back three seasons to the word this project flagged in July, called its most obvious repair, and never removed.

Questions

1. `unan` is crowd-speech vocabulary. **Argue that this is precisely Girard's broken unanimity**, then argue it is a narrative convention. Which needs more assumptions?
2. U4: a gap that clears the margin and misses significance. What do you report?
3. The N3 coding error was caught by a human reading numbers. **Design the check that catches it mechanically.**

Collecting episode 16

Eight rounds, one positive localization result, and every success in the season belonging to the half of the lexicon disclosed in advance as compromised.

This episode leaves the dissent track entirely and settles something that has been open since Season 1 episode 8.

The thing that was flagged and never fixed

`marks` is the bucket that detects the victim's selection: stranger, cripple, blind, orphan, beggar, monster, accursed. And **king**, because Girard says the top of a hierarchy is as marked as the bottom.

The project measured how much of the bucket that one word was.

RECORD	`MARKS` DOMINATED BY	SHARE
round 1	`king`	**54%**
round 2	`king`	**44%**

And `RESULTS-2.md` wrote its own recommendation: **"Fix `marks`. 44% 'king' is not a victim mark, it is a genre signal."**

It was never done. `NEXT.md` filed it under "all real, all deferrable", the four-bucket track was abandoned for the dissent track, and the number sat in the record for three weeks.

The reasoning for deferring it was sound, which is what makes this worth an episode. The four-bucket track had hit its ceiling. Its best possible result was "persecution-shaped texts contain persecution vocabulary", and no repair to `marks` changes that. **Fixing a bucket in an instrument you have correctly abandoned is not obviously a good use of a week.**

> So the word stayed, and every citation of the four-bucket results kept carrying a bucket that was half one word, **with a note in the repository saying so.** The flag was raised, understood, agreed with, and outlived by the thing it was flagging.

The test, pre-registered before the scorer existed

Delete `king` and `kings` from the bucket. **Only `king`.** `queen` and `tyrant` stay, because removing the whole royalty group is a different and larger change, and choosing between the two after seeing which gives a cleaner result is the move this project exists to refuse.

Everything else identical. Same corpus, same matching, same 800-word rule.

The result

`king` is 49.6% of every `marks` hit in the corpus. Higher than the committed 44%. Removing one concept halves the bucket.

And nothing that the project rests on moves.

	BEFORE	AFTER	
named tier separation, lowest HIGH minus highest LOW	+1.69	**+1.56**	survives
MIN non-zero share	90.4%	**89.0%**	survives
mythography vs tragedy gap	21.32%	**21.30%**	unmoved

The falsifiers written in advance were **F1**, the tier separation being destroyed, and **F2**, MIN collapsing. **Neither fired.**

> The worry was real, correctly identified, printed in the record, named the most obvious repair in the project, deferred as real-but-deferrable, and it is **inert.** A number can be alarming and not load-bearing, and **there is no way to know which without running it.**

The device: the prediction that failed, and it is the discovery

The registered prediction was that **tragedy** would lose more than mythography, because drama is full of kings. That is the "genre signal" story.

It runs the other way, and not narrowly.

GROUP	N	MEAN `MARKS` BEFORE	AFTER	FALL
bare mythography	10	3.203	1.126	**−64.9%**
Ovid	11	2.221	1.068	−51.9%
tragedy	12	4.372	2.384	**−45.5%**

Apollodorus, Hyginus and Diodorus are, per word, more about kings than Euripides and Seneca are. Obvious once said, and not predicted. Mythography is largely an account of who ruled where and who their children were. Tragedy has one or two royal figures on stage and a great deal of other language around them.

Think about what a page of Apollodorus is. *X became king of Y. He married Z, daughter of the king of W. Their sons were A and B.* The word is doing structural work in almost every sentence, because a mythographic summary is a list of successions.

Now think about a page of Euripides. Somebody is on stage grieving, or arguing, or a chorus is singing about the sea. **The king is present as a character rather than as a recurring noun**, and the play spends most of its words on things that are not his title.

So "genre signal" survives, and the genre it signals is not drama. It is catalogue.

And notice this is Season 2 episode 4's density confound seen from a new angle, and that **deleting `king` does not touch it** (21.32% to 21.30%). Whatever makes mythography tie tragedy, it is not this word.

The two predictions that were mis-specified, reported as errors

One prediction was built on a misreading. *Oedipus the King* was predicted to lose first place on total rate. **It was never in first place**; `hyginus_fabulae_1` was, and the same page of the record says so. The prediction technically passes and the pass means nothing.

What is worth keeping is the movement nobody predicted:

```
sophocles_oedipus_king   21.22 -> 17.40   -18.0%
seneca_phoenissae        19.90 -> 18.11    -9.0%
```

Oedipus loses twice as much as the other Theban tragedy and drops below it. **The specific worry about that specific play was justified.** The generalisation to the whole bucket was not.

The clinical anchor

Stopping a drug the patient has been on for years, to see if anything happens.

Half the list is inherited. Somebody started it, the indication is not in the notes, and everybody has been renewing it. The only way to find out whether it is doing anything is to stop it and watch.

And the usual answer is nothing happens, which is a real finding and takes a deprescribing clinic to produce, because nobody gets credit for removing a drug that turns out not to have mattered.

The character anchor

Pulling out a wall you were told was load-bearing.

The whole plan was built around it. Everybody says do not touch it. And when it finally comes out, the ceiling does not move, because the load is going through the steel above it and always was.

The house is fine and the plan was wrong, and the only way anyone found out was by taking the wall down.

What would count against this

1. **One word being too small a cut.** `queen` and `tyrant` remain. The royalty group as a whole might be load-bearing where `king` alone is not. **That is a separate round with its own registration** and must not be folded in here.
2. **"Nothing moved" not meaning the bucket is fine.** Deleting the dominant word shows the bucket does not depend on that word. **It does not show the remaining 23 concepts measure a Girardian victim mark.** They have never been validated against attested victims.
3. **Two mis-specified predictions out of six.** Both came from mis-reading the frame that committed numbers sat inside. **Round 2's genre partition exists only as a prose table**, and this round silently substituted a different one.

Handing off

Episode 18 closes the season and adds up what five repairs actually bought, which is less than it sounds and more than it looks.

Questions

1. **Before episode 18:** was this round worth running, given that nothing moved? Answer in a way that would also apply if it had.
2. A partition that lives only in prose gets silently replaced. **What else in this project lives only in prose?**

3. Mythography is more about kings than tragedy is. **What else does that predict about the mythography tier?** Name something checkable.

Collecting episode 17

`king` is 49.6% of the `marks` bucket. Deleting it moved nothing the project rests on, and the prediction that failed was the informative one: the word is proportionally **more** load-bearing in bare mythography than in tragedy.

So the lesson looks settled. **A dominant word is an alarming number and not a problem.** This episode is that lesson being wrong.

The other dominated bucket, flagged in July and never touched

`crisis` is **60% the single word `death`.** Season 1 episode 4 said out loud that nothing in the project ever ran the equivalent test on it, across eight rounds.

Same shape as `king`. Same kind of edit. And there is a reason, written down before scoring, to expect the opposite outcome.

The device: which bucket is MIN actually standing on

MIN takes the smallest of the four bucket rates. So a bucket can only affect MIN when it is the smallest one. Put the means next to the count of texts each bucket binds on:

BUCKET	MEAN RATE	TEXTS WHERE IT IS THE MINIMUM
crisis	2.07	**9**
crimes	**0.95**	**62**
marks	2.90	**2**
violence	4.76	**0**

Read the `marks` row and round 9 stops being mysterious. `marks` was the binding bucket on two texts out of seventy-three. Deleting half of it was guaranteed to be nearly inert, and **that was a fact about arithmetic rather than about the word.**

Now do the projection. `crisis` at 2.07 loses 60% and lands near **0.83**, which is **below `crimes` at 0.95.**

> So `crisis` should stop being the second-smallest bucket and start being the smallest one, across a large part of the corpus. **The pre-registration predicted it would bind on more than thirty texts.**

What came back

It binds on forty-four.

BUCKET MIN BINDS ON	BEFORE	AFTER
crisis	**9**	**44**
crimes	62	29
marks	2	0

The mechanism is confirmed exactly. And then the round fails.

The falsifier fired, and it fired on the measurement

The registered prediction was that MIN would fall. The check used was **the share of texts with non-zero MIN**, which is the check round 2 used, and it came back:

```
MIN non-zero:   90.4%  ->  90.4%
```

Not a tenth of a point. By the letter of the falsifier the whole structural argument is dead.

It is not dead. The measure cannot see the damage.

MIN, MEASURED PROPERLY	BEFORE	AFTER
mean	**0.876**	**0.566**
median	0.818	0.515
texts at exactly zero	7	**7**

A third of the metric is gone. 35.4% of its mean value, and the check that certifies the metric reports nothing.

Work out why, because it is the transferable part and it takes one sentence to see once you have it. **MIN non-zero asks whether the smallest bucket is above zero.** That is a yes-or-no question about a boundary. A text scores zero only if some bucket has **no hits at all**, and deleting `death` did not empty any bucket on any text. Crisis still fires 594 times across the corpus.

So every text that was above zero before is still above zero. **The seven that were at zero are the same seven.** Nothing crossed the line the check watches, and the check has no opinion about anything that does not cross it.

Crisis did not vanish. **It got small**, on forty-four texts, and small is invisible to a test for empty.

> **This is round 2's P6 being a check that cannot fail.** It exists to confirm MIN carries information, it passed at 90.3%, and it is **structurally unable to detect the largest change ever made to the metric it validates.** It moves only at a cliff, and everything that matters happens on the slope.

The comparison the season was built for

DELETION	SHARE OF ITS BUCKET	MIN MEAN BEFORE	AFTER	FALL
`king`, round 9	49.6%	0.876	0.873	**0.3%**
`death`, round 10	**61.9%**	0.876	**0.566**	**35.4%**

Two words, comparable shares of their buckets, roughly a hundredfold difference in what removing them does. **Not because one word matters more.** Because of where its bucket sits relative to the other three.

That is worth stating as a rule, because it generalises past this instrument.

> **A component's importance is not its size. It is whether it is the one your metric is currently standing on.** `marks` was 2.90 per thousand and mattered almost not at all. `crisis` was 2.07 and mattered enormously. The larger bucket was the irrelevant one.

And notice what that does to the audit you would have run. Ranking the four buckets by how dominated they are by a single word puts `crisis` at 60% and `marks` at 49.6%, which makes them look like two instances of one problem. **They are not the same problem and the ranking cannot tell.** You have to know which bucket the metric binds on, and nothing in the record recorded that until this round computed it.

And the thing that survived both

	TIER SEPARATION
before any deletion	**+1.69**
after deleting `king`	+1.56
after deleting `death`	**+1.41**

The named tier separation, round 2's only cleanly passing prediction, loses **9.4%** across a deletion that removed a third of MIN. **Two independent amputations and it holds**, which is the strongest thing anyone has said for that result.

The clinical anchor

A pass-fail threshold on a test that reports a value.

The lab flags potassium as abnormal outside a range. A patient drifting from 4.0 to 3.6 is flagged as normal at both ends, and the flag has told you the truth twice while the thing you cared about moved.

The flag is not wrong. It is answering the question it was given, and the question was a boundary rather than a trend.

The character anchor

A fuel gauge that only has a warning light.

The light is honest. It comes on when the tank is nearly empty, and it has never lied to anybody. Drive from full to a quarter and it stays off the entire way, correctly.

You have lost three quarters of your fuel and the instrument's report is unchanged, and nobody would call the light broken, because the light was never a gauge.

What would count against this

1. **The 35.4% not mattering.** MIN falling by a third changes no verdict in the record, because nothing in rounds 1 to 3 was near a threshold. **True, and it is the reason this is a finding about the check rather than about a result.**
2. **P6 being fit for its stated purpose.** It was written to show MIN is not saturated at zero on short texts, and it does that. **The objection is that it is quoted as though it validates MIN generally**, including in this series.
3. **My prediction being the actual failure.** It was. D2 and D5 both used the non-zero share, so **the round failed by inheriting exactly the defect it found**, and that is reported rather than tidied.

Handing off

Episode 19 closes the season and adds up six repairs, of which two were inert, one was mis-measured by its own author, and one worked.

Questions

1. **Before episode 19:** `violence` binds MIN on **zero** texts. Predict what deleting `blood` from it would do, and say why you are now confident.
2. Design the replacement for P6. What would it measure, and what would make it fail?
3. Every "MIN non-zero" figure in the record is weaker than it reads. **Go and find the ones this series has quoted**, and decide which need a caveat.

Collecting episode 17

`king` is half the `marks` bucket and deleting it changes nothing load-bearing. The prediction that failed was the informative one, and the genre it signals is catalogue rather than drama.

The season, added up

ROUND	THE REPAIR	VERDICT
4	a fifth bucket for dissent	**FAIL**, destroyed by its own registered controls
5	rebuild the lost null script	**UNSTABLE**, then identified by two bystander numbers
6	a null that matches the instrument	**PASS**, 4 of 4
7	ask where the words sit	**FAIL**, under-powered by its own check
8	ask again, violence held fixed	**U1 PASS**, three fails, N3 fired
9	delete `king`	**inert**
10	delete `death`	**F1 fired, on the measure rather than the mechanism**

Two passes in seven. And the two passes are in the same place.

Before the pattern, notice what the season did that Seasons 2 and 3 could not. Every round here was a repair of a specific, named, previously-recorded defect. Round 6 repairs round 4's null. Round 8 repairs round 7's. Round 5 repairs a missing file. Rounds 9 and 10 repair a flag raised in July, and **round 10 is the only one that came back with something the record did not already suspect.**

> **This is a project reading its own record and acting on it**, which is the behaviour everybody claims and almost nobody can demonstrate, because it requires the record to have been written honestly enough to act on. Seasons 2 and 3 are what paid for this season.

The one sentence this season exists to deliver

> **Everything that worked belongs to `unan`.** Division, some said, others said, not all, spare him, withheld consent. **The `core` half has never carried anything, on any measure, in any round.**

Season 3 episode 16 read that `core` list out when the bucket was still an idea, and it was the half that sounded like Girard. **Every word in it is a word about a victim's innocence**, which is what anybody would build first and what anybody would defend in an interview.

Four rounds, two translations, three nulls, and it carries nothing. Not a weak signal. **Nothing**, and on round 8 it points backwards.

And the half that does carry it was disclosed as compromised before it was built, because the author already knew that John repeats a division formula.

The device: the *Bacchae*, four pre-registered tests

Girard's single most important persecution myth.

ROUND	OPERATIONALIZATION	RESULT
2	whole-text rate, four buckets	rank **35 of 72**
3	`peak_min` against a word-shuffle null	**p = 0.66**
7	dissent-to-violence distance, uniform null	**p = 0.888**
8	co-location, violence held fixed	**p = 0.414**

Four operationalizations, four failures. Season 1 episode 15 planted this table and told you where it detonates. This is it.

And Season 1 episode 15 also found that the *Bacchae* **does** contain dissent, from Cadmus and Tiresias, **before and after the killing and never during**. Round 8's statistic is measuring exactly that gap and reporting 0.4802 against a baseline of 0.5.

> **The instrument and the close reading agree.** That is the strongest moment of convergence in the project, and what they agree on is that Girard's central example does not behave the way his theory says it should.

What this season answers, from Season 3's question

Season 3 episode 19 asked you to write down: **wrong, or blind?** Four rounds of failure with a competent explanation for each one.

The answer is neither, and the season shows why the question was badly formed.

The project was not wrong: the phrase separation is real, it survived a matched null at 0 of 2000 and a 290-year translation swap. The project was not blind: it built the power falsifier, it fired, and it was reported.

What actually happened is narrower and worse. The instrument works, and the half of it that works is the half that measures crowd speech rather than the questioning of guilt. **Nobody chose that.** It was disclosed as a risk in advance and then it came true, four times, and each time it was reported.

Why "worse" is the right word, given that nothing was hidden. **Wrong is recoverable and blind is fixable.** A wrong theory gets replaced. A blind investigator gets a checklist. Both have a next step.

This has no next step of that kind. The disclosure was made, the risk was named before it materialised, the sensitivity analyses were registered in advance, they fired, and they were reported in the headline as the pre-registration demanded. **Every single procedure worked**, and what they produced was a result whose meaning nobody can settle, sitting in the half of an instrument that was flagged as compromised before it was built.

> **There is no procedural failure here to point at and repair.** The project did everything it said it would do, and arrived somewhere it cannot interpret. That is a harder position than being wrong.

What is still missing, said plainly

- **`violence` is dominated by `blood` and binds MIN on zero texts**, so a deletion round on it is guaranteed inert before anybody runs it. That is now predictable rather than surprising, which is what episode 18 bought.
- **A replacement for P6.** The check on MIN is known to be blind to magnitude and nothing has been written to replace it.
- The `core` half has never been deleted, only reported as carrying nothing.
- Four Gospel texts is the whole Gospel tier, and U4 failed on it.

The clinical anchor

A treatment that works, for a reason other than the one it was given for.

The patient improves. The drug is real, the improvement is real, and the mechanism everybody wrote in the notes is not the mechanism.

Nothing about that is fraud and it changes what you can do next, because the next patient is selected on the wrong indication.

The character anchor

A detective who solves the case and convicts on the wrong evidence.

The right person is in prison. The confession, the motive, the timeline all hold. And the fingerprint that closed it was contaminated, so **the case is sound and the method that produced it cannot be used again.**

What would count against this

1. **`unan` being Girardian.** Stated as the strongest objection in the season, in episode 16, and it is not answered here. **A crowd that says "some said, others said" is a crowd that is not one voice.** Season 7 has to take it seriously rather than filing it as contamination.
2. **Two passes in six being a bad rate.** Or a normal one. **Nobody in this project has said what rate a healthy programme has**, and without it "two of six" is a number with no scale.
3. **The season repairing things nobody had shown were broken.** Round 9 is the clearest case: a repair to a fault that was inert. **How many of the other five were the same?**

Handing off

Season 5 turns the instrument on the Bible itself, and the first thing it has to face is the objection this season kept deferring: whether four texts about one week, written by the victim's own partisans, can be compared to anything at all.

Questions

1. **Two deletions, opposite outcomes, one explanation.** State it in a sentence without using either word.

2. Two passes in six rounds. **What rate would a healthy research programme have?** You need this number before Season 5.
3. The instrument and the close reading agree about the *Bacchae*. **Is that convergence, or two readings by the same person?** Say what would tell them apart.